LEARNER GUIDE · COURSE SLIDES

Agentic AI Automation with n8n

WSQ Course Code: TGS-2023035977

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v47 · 8 Aug 2026

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator will display the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD — specialises in AI, automation and software engineering.

DELIVERS
WSQ courses on AI agents, automation (n8n) and app development.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with automation or AI tools (if any).
- What you want to automate after this course.

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HOUSEKEEPING

Ground Rules

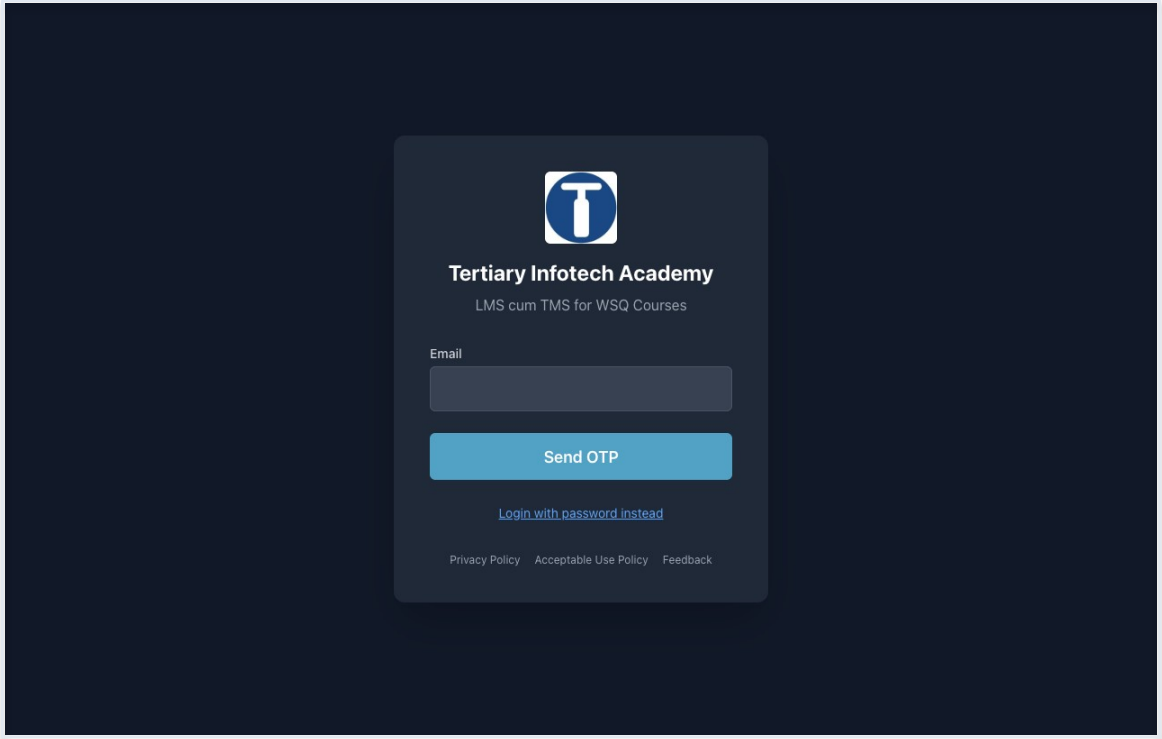
1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Download Course Material

- Go to lms-tms.tertiaryinfotech.com.
- Enter your registered email and click Send OTP.
- Key in the one-time password from your inbox to log in.
- Open this course to download the slides and Learner Guide.
- Keep them open — the assessment is open book.
- Attendance and assessment submission also happen here.



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GET THE WORKFLOWS

Download the Course Flows (GitHub)

1. Open the course repo: github.com/tertiarycourses/TGS-2023035977-Agentic-AI-Automation-with-n8n
2. Code ▾ → Download ZIP (or git clone) — all workflows are in the labs/ folder.
3. In n8n: Workflows → Add workflow → ... → Import from File → pick the activity .json.
4. After importing, re-select YOUR OWN credentials on each node, then Save & Publish.

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SCHEDULE

Lesson Plan — 3 Days, 9:30am–6:30pm

Day 1 — Automate, then add AI

- Topic 1: n8n basics — triggers, actions, data
- Activities 1, 2, 3a, 3b — forms, storage, logic
- Topic 2: AI Agents — model · memory · tools · instructions
- Activities 4a, 4b — Telegram agent + Data Table tool

Day 2 — Connect it to the world

- Topic 3: Webhooks — Activity 5 (website chatbot)
- Topic 4: APIs & HTTP — Activity 6 (day-trading agent)
- Topic 5: RAG — Activities 7a, 7b (grounded agents)

Day 3 — Govern, then prove it

- Topic 6: Security — human-in-the-loop + guardrails
- Activities 8a, 8b, 8c — the HR Service Portal
- Topic 7: Mini capstone + presentations
- Written Assessment (SAQ) 1 hr · Practical (PP) 1 hr

Each day runs 9:30am–6:30pm — 8 instructional hours with a 1-hour lunch; tea breaks counted within.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1. LO1: Build automated workflows in n8n using triggers, actions, nodes and flows.
2. LO2: Design AI agents with LLMs, memory and tools, triggered from chat / Telegram.
3. LO3: Apply Retrieval-Augmented Generation (RAG) to ground agents in your documents.
4. LO4: Integrate external systems via webhooks, APIs and HTTP requests.
5. LO5: Apply human-in-the-loop and guardrails to secure agentic automations.

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COURSE JOURNEY

Twelve Connected Labs

1. Act 1 — Form → Email: your first automation, on a real flyer.
2. Act 2 — Data Table: store every submission, not just forward it.
3. Act 3a/3b — IF logic: route Yes/No to storage or a thank-you.
4. Act 4a — Telegram AI agent: model + memory + instruction.
5. Act 4b — Agent + tool: answers HR questions from a Data Table.
6. Act 5 — Webhook: your agent behind a public website chatbot.
7. Act 6 — APIs: live candles + news → Buy / Sell / Hold calls.
8. Act 7a — RAG: upload a PDF, ask the bot in Telegram.
9. Act 7b — RAG at scale: 20 brochures in a real vector DB.
10. Act 8a–8c — HR portal: approval gate, dashboard, guardrails.
11. Capstone — design and demo your own end-to-end automation.

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BEFORE THE ASSESSMENT

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

- Written Assessment (SAQ) — 1 hour.
- Practical Performance (PP) — 1 hour.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A mini capstone project is presented on Day 3.
- An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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01

TOPIC 1

Workflow Automation with n8n

What n8n is · triggers · data-processing nodes · JSON and expressions · executions

Applied in Activities 1, 2, 3a and 3b

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TOPIC 1 · WORKFLOW AUTOMATION

What Is n8n?

A workflow automation platform: connect apps, data and AI on one visual canvas — no-code to full-code.

1 Build visually

Drag nodes onto a canvas and wire them together. 400+ app integrations, plus HTTP and Code nodes when no integration exists.

2 Run anywhere

Free cloud trial at n8n.io, or self-host with Docker. Webhooks, schedules and chat triggers work the same in both.

3 AI-native

AI Agent and LangChain nodes are built in — attach any LLM (OpenAI, Gemini, Claude), memory, tools and vector stores.

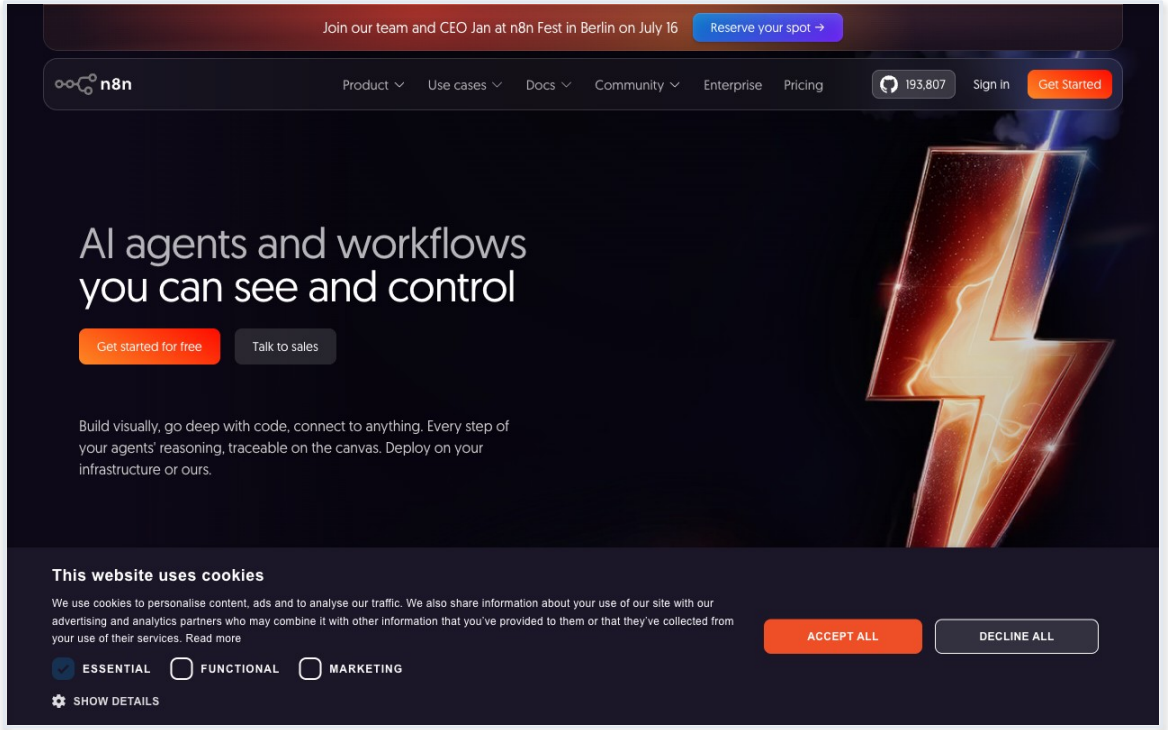
One canvas to connect apps, data and AI — automation and agents without gluing code by hand.

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THE PLATFORM

n8n.io

- n8n is an open, fair-code workflow automation platform.
- Free cloud trial at n8n.io, or self-host with Docker.
- Same visual editor in both.
- This course: start on the cloud trial, keep the Docker option for after.



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What Automation Removes

Automation is not 'the same work faster' — it removes the hand-offs where a person re-types what another system already knows.

MANUAL

Form filled in ▶ Someone reads it ▶ Re-typed into Excel ▶ Someone remembers to reply

AUTOMATED

Form submitted ▶ Flow triggered ▶ Row written ▶ Reply sent

Consistency

The same input produces the same output, every time, at 3am.

Traceability

Every run leaves an execution history someone can read a year later.

Capacity

People move up to judgement calls; the machine does the re-typing.

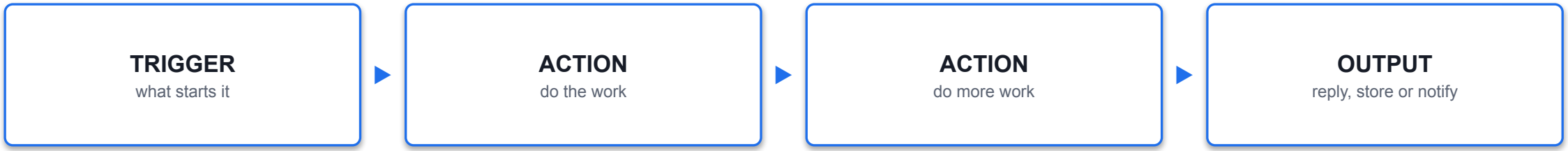
Automate the hand-offs first — that is where time and errors live.

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TOPIC 1 · WORKFLOW AUTOMATION

Every Workflow Is the Same Shape

Trigger first, then actions in order, with data flowing left to right. Only the nodes change.



1 One trigger

A workflow has exactly one entry point. Change the trigger and you have a different flow — the actions may not change at all.

2 Actions are ordered

Each action runs after the one before it and can read the outputs of every node before it.

3 Data flows as JSON

Every node passes a list of JSON items to the next — the same structure in every lab.

Trigger, actions, output. Learn this once in Activity 1 and every later lab is a variation.

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Trigger Nodes — What Is Allowed to Start a Workflow

The trigger you choose is a statement about who or what may start the process.

1 Manual Trigger

Click Execute Workflow — for building and testing, and for one-off jobs like the Activity 7b ingestion run.

2 Schedule Trigger

A clock starts the flow — cron or interval. Reports, reminders, nightly syncs.

3 Form Trigger

n8n hosts a web form for you; each submission starts a run. Activities 1–3.

4 Webhook

External systems call a URL you expose. Websites and apps — Activities 5, 7b, 8.

5 Chat / Telegram Trigger

Each chat message starts a run — n8n's chat UI or your Telegram bot. Activities 4, 6, 7a.

6 App & Email Triggers

On new Gmail, new Sheet row, new Drive file, new IMAP email — react to changes in apps you already use.

Pick the trigger by asking: who or what should be allowed to start this process?

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Triggers Used in This Course

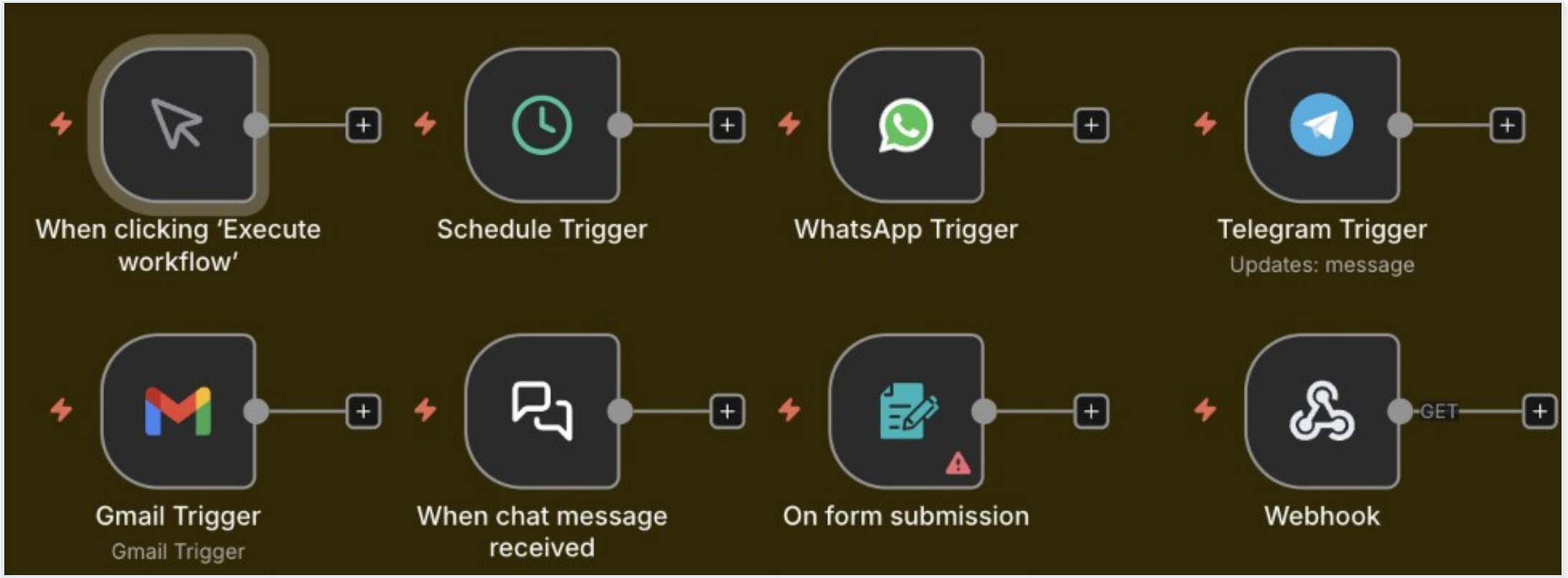
Every lab's trigger, and why that trigger is the right front door.

Trigger	Where	Why it fits
Form Trigger	Activities 1, 2, 3a, 3b	A visitor RSVP must start the flow
Telegram Trigger	Activities 4a, 4b, 6, 7a	Each bot message is one conversation turn
Webhook — POST	Activities 5, 7b, 8a, 8c	A website sends data and waits for a reply
Webhook — GET	Activity 8b	The dashboard pulls JSON from a URL
Manual Trigger	Activity 7b ingestion	A human runs the document load once

Schedule and app triggers follow the same pattern — swap the front door, keep the flow.

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Trigger Nodes on the n8n Canvas



n8n trigger nodes: Manual · Schedule · WhatsApp · Telegram · Gmail · Chat · Form · Webhook

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Data-Processing Nodes — The Workhorses

Between the trigger and the output, these nodes shape, route and transform the data.

1 Edit Fields (Set)

Add, rename or reshape fields — the node you reach for first when data is the wrong shape.

2 IF / Switch

Branch the flow: IF gives true/false paths; Switch gives one path per case.

3 Filter

Let only matching items continue — drop the rest of the list.

4 Merge

Combine two branches — append rows, match by key, or choose one side.

5 Split Out / Aggregate

Turn one item holding a list into many items — and back again.

6 Code

JavaScript or Python for the 5% a built-in node cannot do.

7 Loop / Wait

Process items in batches; pause a flow for time or an event.

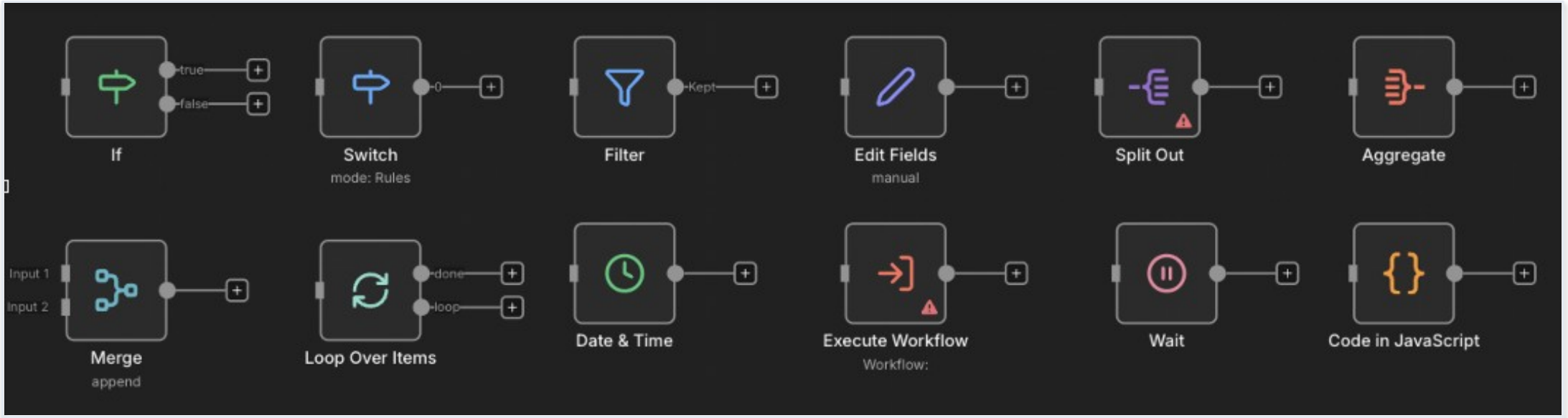
8 HTTP Request

Call any API on the internet — the universal escape hatch.

Most flows are: trigger → reshape (Set) → decide (IF) → act (app node). Reach for Code last.

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Data-Processing Nodes on the Canvas



If · Switch · Filter · Edit Fields · Split Out · Aggregate · Merge · Loop · Date & Time · Execute Workflow · Wait · Code

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Where the Data Goes — Storage & App Actions

Action nodes that write the result somewhere useful.

1 Data Table

n8n's built-in spreadsheet — instant, no credential. Trial Data Tables are NOT permanent; use for class demos.

2 Google Sheets / Excel

A real spreadsheet you keep — OAuth2 credential, Append Row. The persistent upgrade in Activity 3b.

3 Gmail / Outlook

Send (and in Activity 8a, Send and Wait for Response) — email as an output and as an approval gate.

4 Vector Store

Store embeddings for RAG — in-memory for demos, Pinecone / Supabase / Qdrant for production.

Anything you must keep goes to external storage — trial Data Tables vanish with the trial.

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JSON — How Data Travels Between Nodes

Every node passes a list of items; each item is a JSON object of key/value pairs.

- JSON (JavaScript Object Notation) is a simple text format for data.
- Values can be text, numbers, true/false, lists [ ] or nested objects { }.
- Most APIs send and receive JSON — so does every n8n node.
- Click any node to inspect its input and output JSON.

ONE FORM SUBMISSION AS JSON

```
{
  "Name": "Alice Tan",
  "Email": "alice@corp.com",
  "Attending": "Yes",
  "Pax": 2
}
```

Read any value later with an expression: {{ $json.Name }} → "Alice Tan"

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Expressions — The Token, Not the Text

An expression references a value that will exist at run time — it is resolved fresh on every run.

Trigger runs
a submission arrives

▶

Outputs exist
Name, Email, Attending…

▶

Later node references them
{{ $json.Name }}

▶

Value arrives at run time
different every run

✓ Mapped with the picker

Drag the field from the input panel — n8n writes the expression for you and shows a live preview of the resolved value.

✗ Typed as plain text

Type 'Name' as literal text and every email says 'Name'. If the path is wrong the value resolves to empty — not to an error.

Use expressions in any field — subject lines, URLs, conditions. Drag, don't type.

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Test vs Production Executions

The same workflow runs in two modes — know which one you are looking at.

TEST — in the editor

- Click Execute Workflow to run once.
- Data is shown live on the canvas, node by node.
- Pin a node's output to freeze it while you build downstream.
- Perfect for building — nothing needs to be Active.

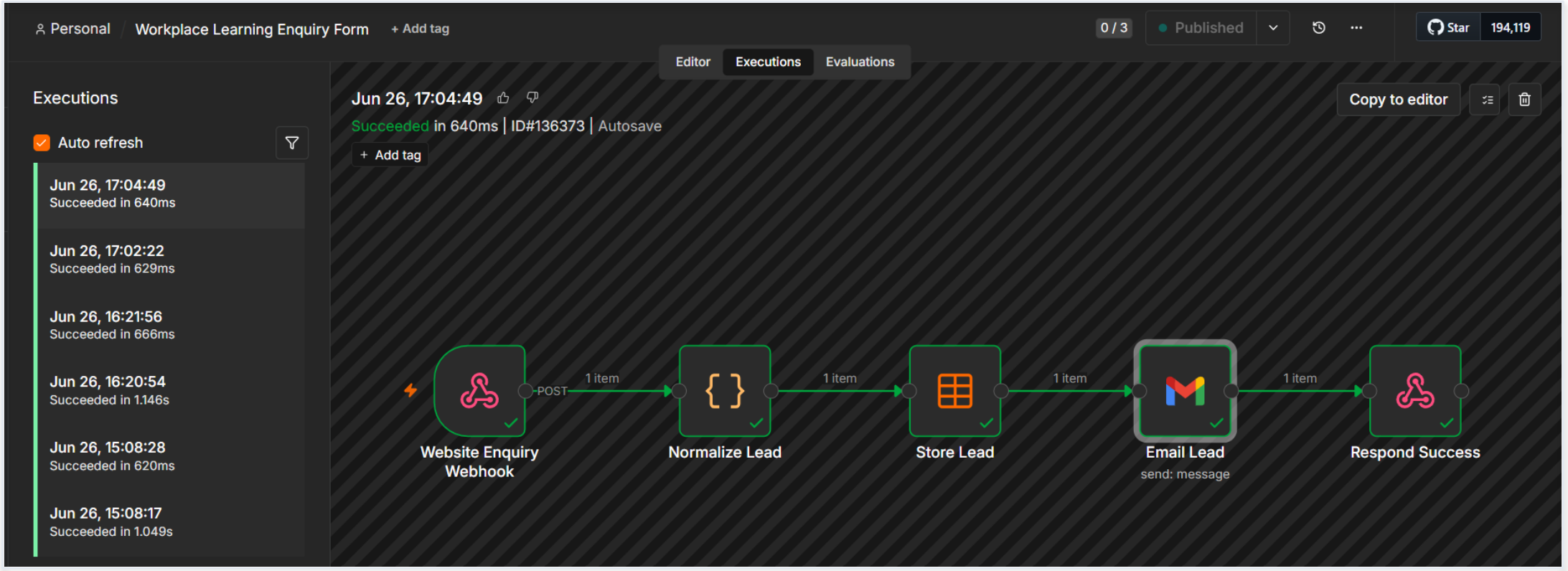
PRODUCTION — Active toggle ON

- Runs automatically when the trigger fires.
- Nothing appears on the canvas — runs are logged in Executions.
- Every run records status, timing and full data per node.
- This is what your users experience.

The Executions list is the honest account of what happened — debug from there, not from memory.

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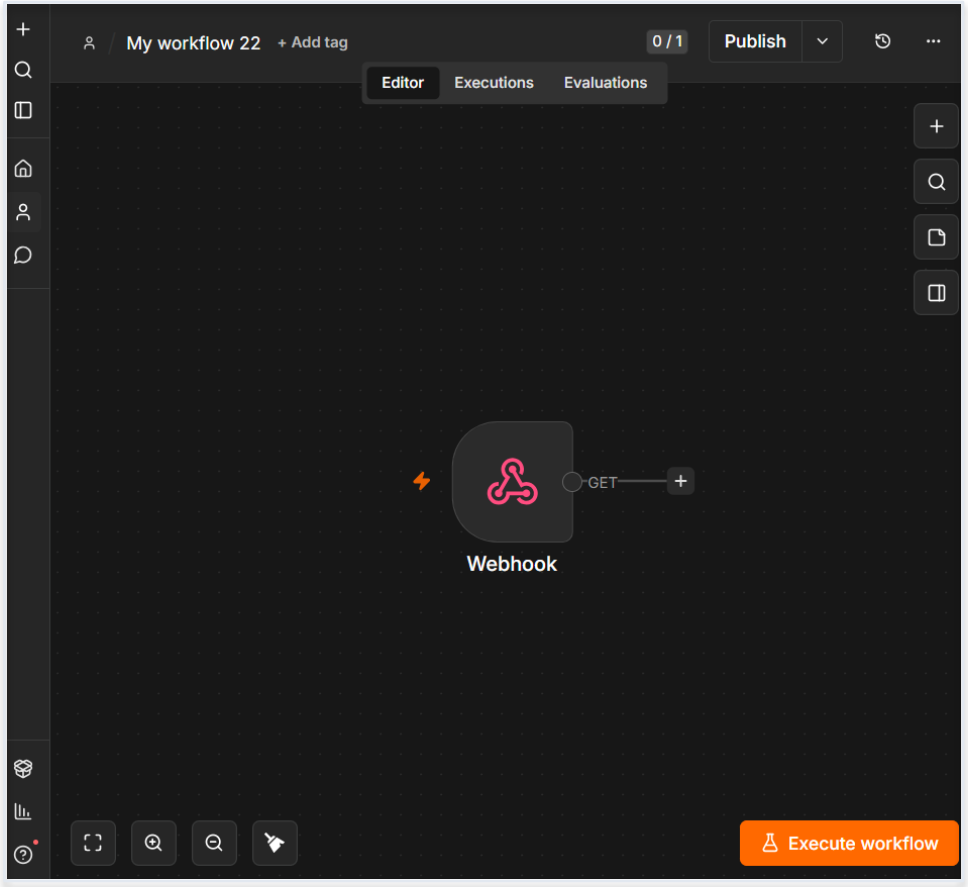
Execution History



Executions tab — each run shows status, duration, ID and the full data at every node

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The n8n Editor



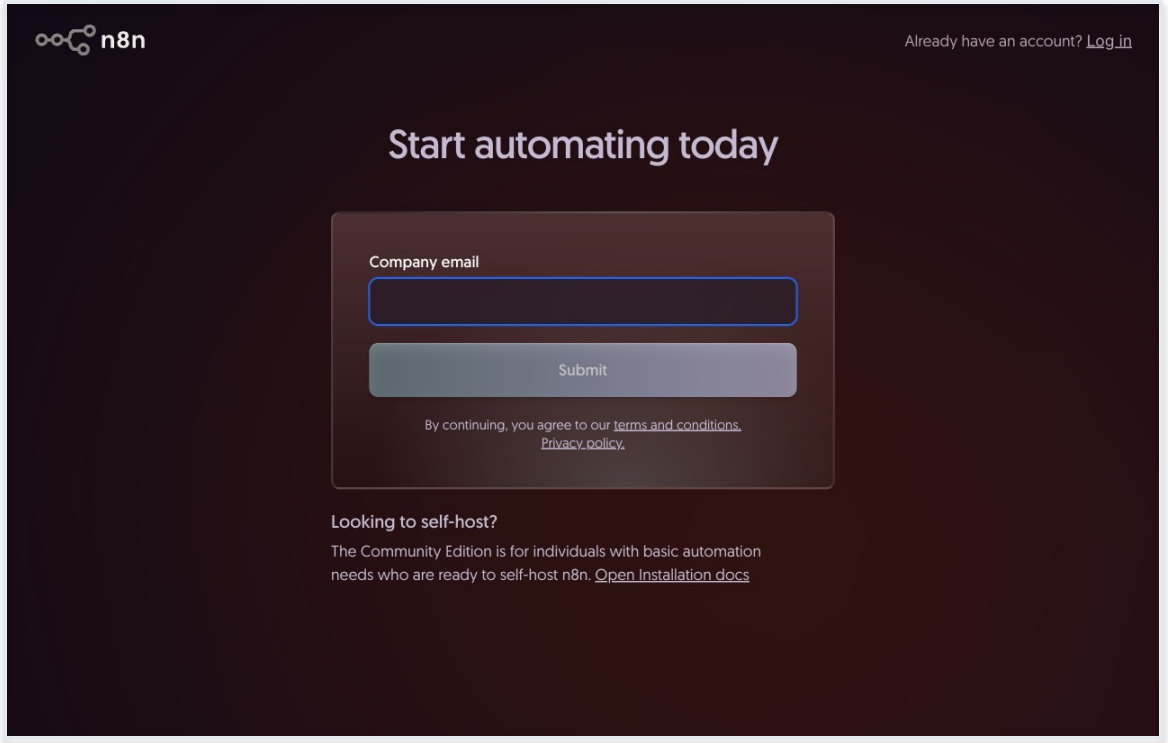
n8n canvas: left sidebar navigation · centre canvas · node toolbar · Editor / Executions / Evaluations tabs

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YOUR LOGIN · CLOUD INSTANCE

Sign Up for n8n Cloud

- Go to app.n8n.cloud/register in your browser.
- Enter your company email and click Submit.
- Open the verification email from n8n and click the link to confirm.
- Set your password and complete the short profile questions.
- n8n creates your workspace at <name>.app.n8n.cloud and opens the editor.
- The free trial runs 14 days — no credit card needed.



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Credentials — Connect Your Accounts Once

Nodes reference credentials by name; set them up once under Credentials → Add credential.

1 Email

Gmail or Outlook via OAuth2 — used from Activity 1 onward.

2 LLM

OpenAI / Google Gemini API key — powers every AI agent from Activity 4.

3 Telegram

Bot token from @BotFather — the chat front door for Activities 4, 6, 7a.

4 Google Sheets

OAuth2 to your Google account — persistent storage in Activity 3b.

Imported workflows keep credential NAMES only — always re-select your own after importing.

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TOPIC 1 · LAB WORKFLOW

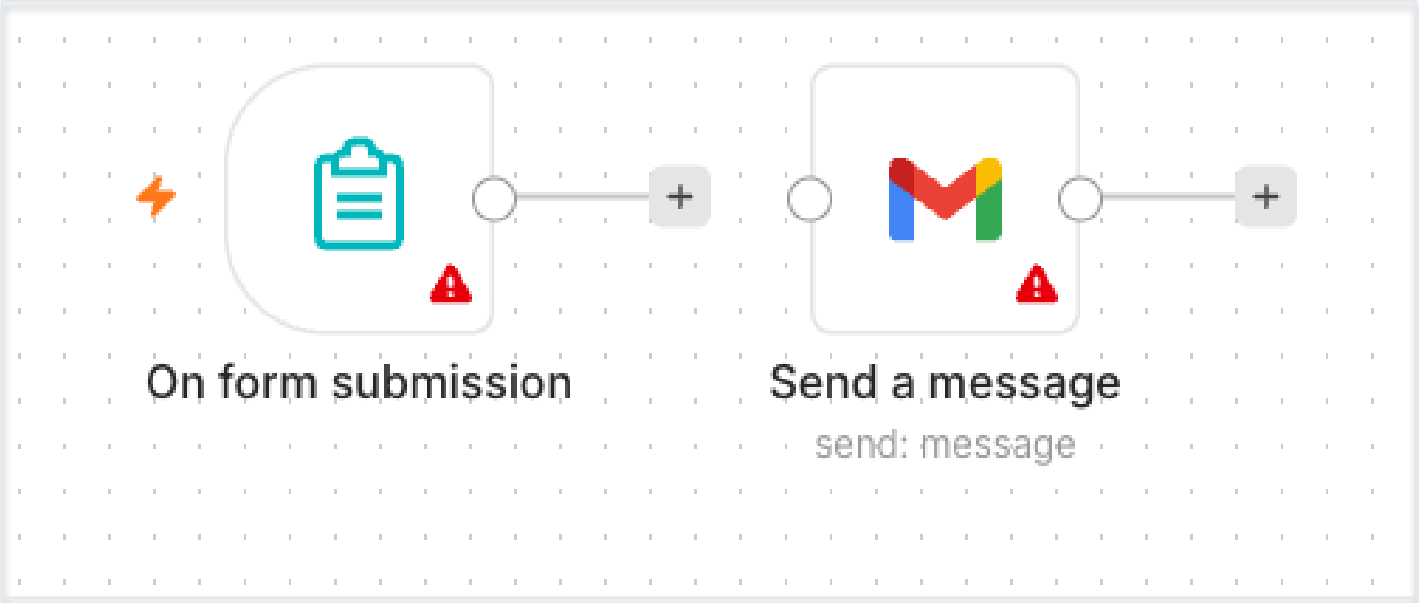
Activity 1 — Flyer with QR Code (Form → Email)

Collect a visitor's RSVP with a hosted n8n Form and email it to the admin — then put the form's QR code on a real event flyer.

Form Trigger
visitor submits the RSVP

▶

Gmail — Send
admin receives the details



ACT 1

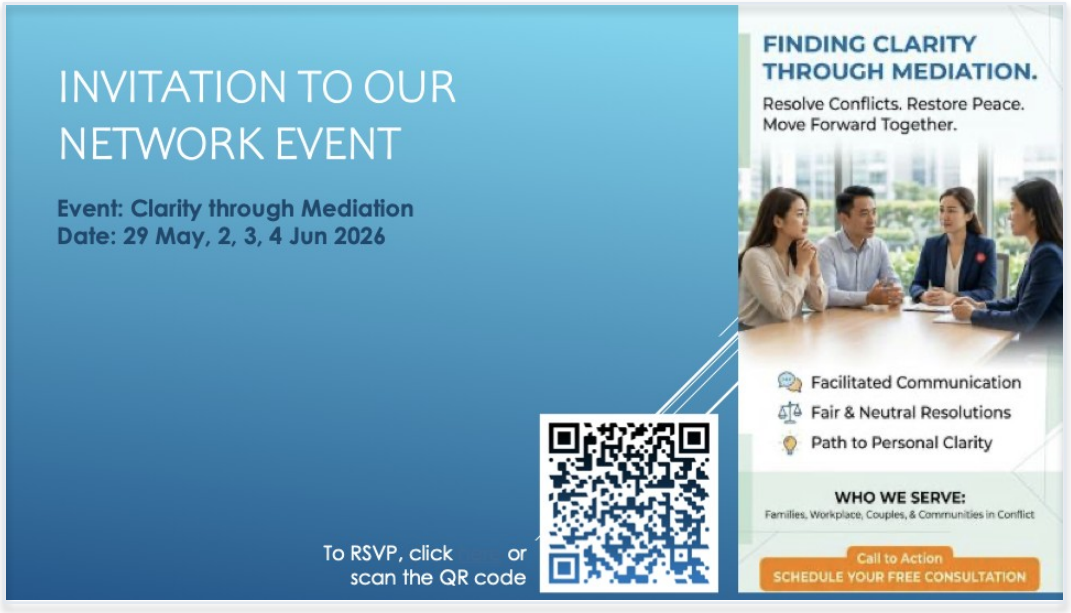
Flyer with QR Code (Form → Email)

Form Trigger → Gmail · Gmail OAuth2 credential · full steps in the Learner Guide

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ACT 1 · GROUP ACTIVITY

Sample Flyers from Past Students



Network event



Bowling party

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TOPIC 1 · LAB WORKFLOW

Activity 2 — Capture Submissions in a Data Table

Extend Activity 1 so every submission is also saved — your first taste of storing data, not just forwarding it.

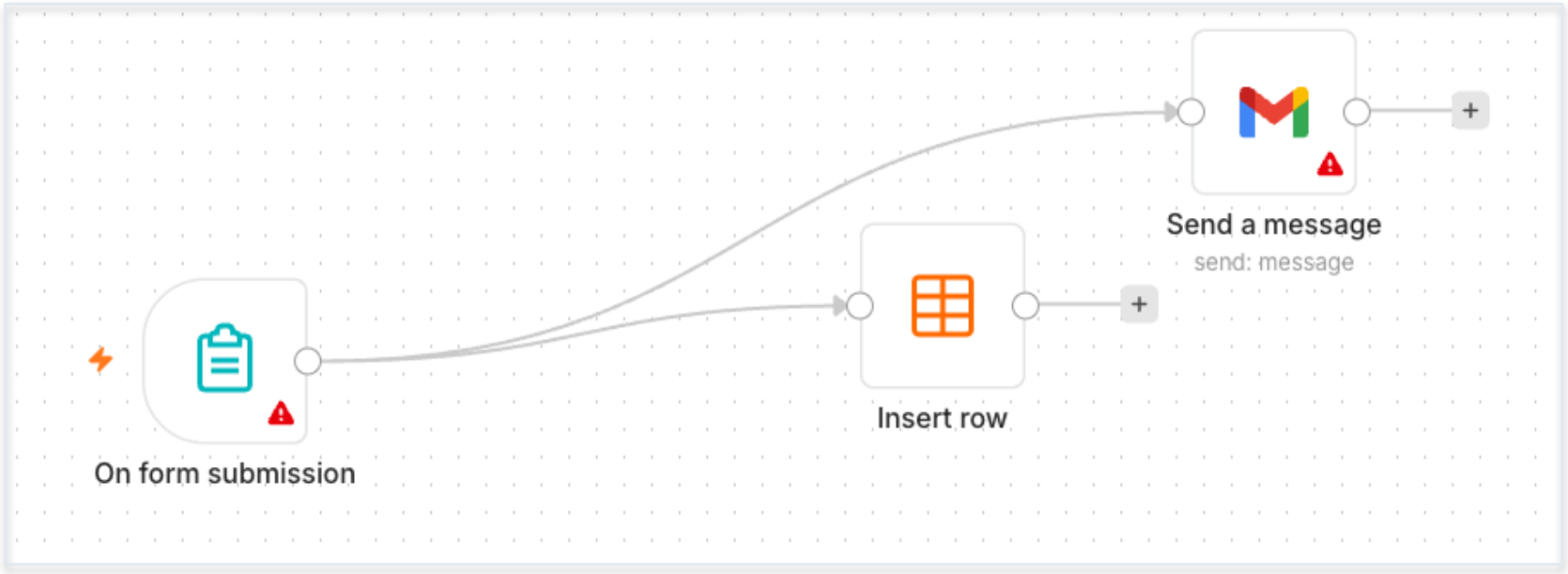
Form Trigger
same RSVP form

▶

Data Table — Insert Row
every submission saved

▶

Gmail — Send
admin still notified



ACT 2

Capture Submissions in a Data Table

Form Trigger → Data Table + Gmail · columns mapped with {{ $json.… }} expressions · steps in the Learner Guide

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TOPIC 1 · LAB WORKFLOW

Activity 3a — Conditional Response (Data Table)

Add decision-making: attending guests are saved; everyone else gets a polite thank-you email.

Form Trigger
+ Attending? Yes/No field

▶

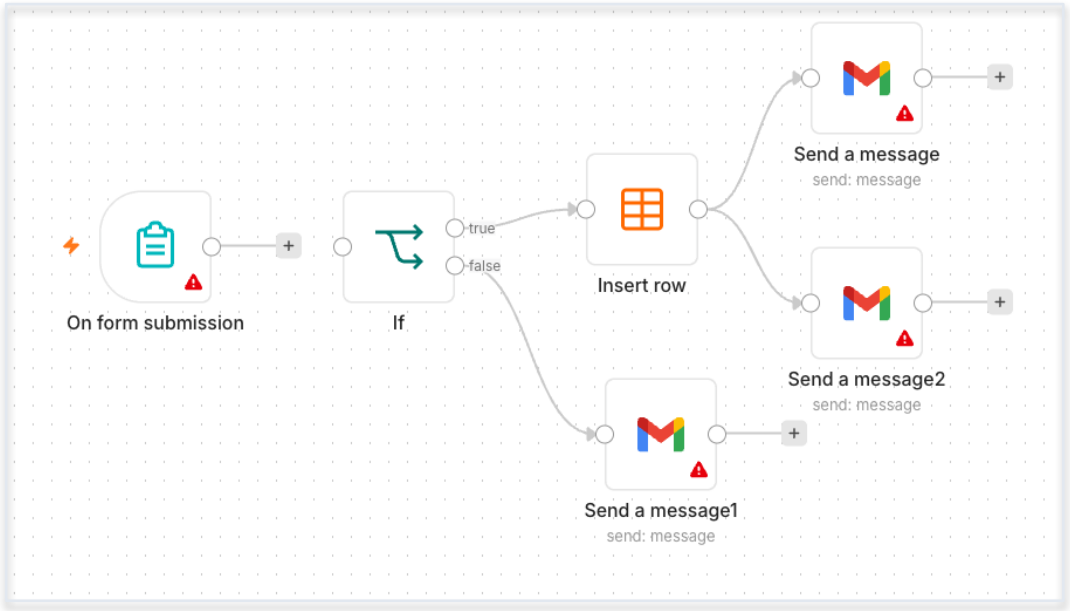
IF
Attending = Yes ?

▶

Data Table
YES → save the RSVP

▶

Gmail
NO → thank-you email



ACT 3a

Conditional Response (Data Table)

Form → IF → Data Table / Gmail · both branches must be built · steps in the Learner Guide

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TOPIC 1 · LAB WORKFLOW

Activity 3b — Conditional Response (Google Sheets)

Same logic, persistent storage: the YES branch appends to a real Google Sheet you keep after the trial.

Form Trigger
same form

▶

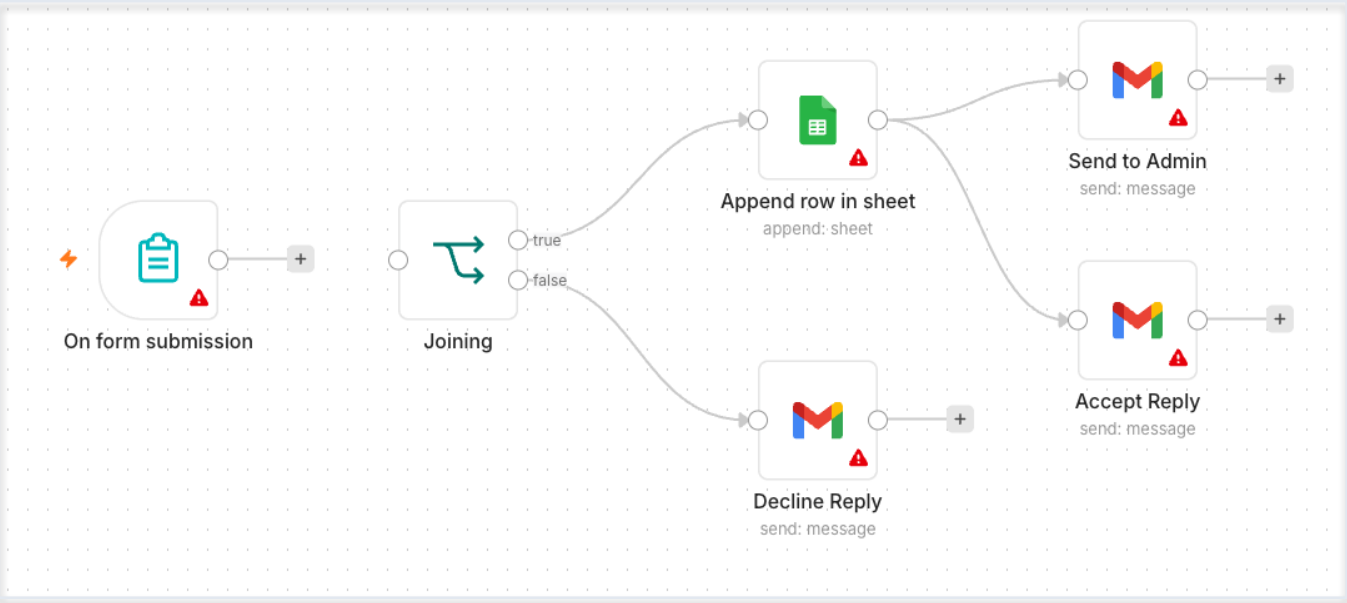
IF
Attending = Yes ?

▶

Google Sheets — Append
YES → a spreadsheet you keep

▶

Gmail
NO → thank-you email



ACT 3b

Conditional Response (Google Sheets)

Form → IF → Google Sheets / Gmail · Google Sheets OAuth2 · steps in the Learner Guide

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Lunch Break

1 hour · see you at 2:00 pm

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02

TOPIC 2

AI Agents

Generative vs agentic AI · model, memory, tools, instructions · model parameters · MCP

Applied in Activities 4a and 4b

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Generative AI vs Agentic AI

GENERATIVE AI — answers

- You ask; it writes one answer.
- No memory of your systems, no actions.
- Quality depends on what you paste into the prompt.
- Example: ChatGPT summarising a document.

AGENTIC AI — acts

- Works toward a goal, not just one reply.
- Calls tools: looks up data, calls APIs, writes records.
- Holds a conversation with memory.
- Example: a bot that checks the Employees table and answers.

Generative AI produces content. Agentic AI decides and does — which is why it needs governance.

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Components of an AI Agent

Four parts, one hub: the agent node wires them together — and only some are under your direct control.

MODEL

The LLM 'brain' — interprets language, reasons, chooses the next step. gpt-4.1-mini in this course.

MEMORY

Keeps the conversation — follow-up questions still make sense. Window buffer per chat/session.

THE AI AGENT

plans · decides · calls tools

TOOLS

Abilities it may use: Data Table lookups, vector stores, HTTP calls, sub-workflows.

INSTRUCTIONS

The system message — identity, scope, rules and refusals. Always in force.

Model + Memory + Tools + Instructions — every agent in this course is exactly these four parts.

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The Model — Popular LLM Providers

Any chat model plugs into the same AI Agent node — swap the brain, keep the agent.

Provider	Models	Notes
OpenAI	gpt-4.1, gpt-4.1-mini, gpt-4o	Course default: gpt-4.1-mini
Google	Gemini 2.x Flash / Pro	Used for embeddings in Activity 7
Anthropic	Claude (Opus, Sonnet, Haiku)	Strong reasoning and safety
Meta	Llama 3.x	Open weights — self-hostable
Mistral	Mistral / Mixtral	Open weights — EU option

Pick by task: small fast models for classification and guardrails, larger models for reasoning.

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Model Parameters — Temperature and Friends

Settings on the chat-model node shape how the model answers — the most important is temperature.

1 Temperature 0 – 0.3

Deterministic and repeatable. Use for classification, guardrails, data extraction — automation that must give the same decision twice.

2 Temperature ≈ 0.7

Balanced — natural-sounding chat with some variety. The default for customer-facing assistants.

3 Temperature 1.0 +

Creative and surprising. Marketing copy and brainstorming — never for decisions.

1 Top-p

Alternative sampling control — limits choices to the most probable tokens. Tune temperature OR top-p, not both.

2 Max tokens

Caps the length of the reply — protects cost and keeps chat answers short.

3 Frequency / presence penalty

Discourage repetition and nudge new topics — useful for long generations.

For business automation keep temperature LOW — the same input should produce the same decision.

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TOPIC 2 · AI AGENTS

Memory — Why the Agent Remembers

Without memory every message is a stranger; with it, 'what about tomorrow?' makes sense.

Msg 1: 'Any leave left?'
agent answers: 12 days

▶

Stored in memory
keyed by chat / session ID

▶

Msg 2: 'What about Bob?'
agent knows the topic is leave

▶

Window slides
only the last N turns kept

Simple (buffer window) memory

Keeps the last N exchanges of THIS conversation — n8n's default, used in all course agents.

Session key

Memory is per chat ID — two Telegram users never see each other's context.

Memory is conversation context — it is NOT long-term knowledge. Facts live in tools and documents (RAG).

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Tools — What the Agent May Do

Attach tools to the agent's Tool input; the model decides when to call one, based on your description.

1 Data Table / Sheets tool

Look up structured records — the Employees table in Activity 4b.

2 Vector Store tool

Retrieve document chunks for grounded answers — the knowledge_base in Activity 7.

3 HTTP Request tool

Call any external API mid-conversation — prices, weather, CRM.

4 Sub-workflow tool

Wrap an entire n8n workflow as one callable tool — your own logic, on demand.

The tool DESCRIPTION is how the agent knows when to call it — write it for the model, not for a developer.

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Instructions — The Four Parts That Constrain

The system message is always in force. Every good agent instruction in this course states four things.

1 Identity

Who the agent is and what it must never do. 'You are HR Buddy, the HR assistant for staff of Acme Pte Ltd.'

2 Source rules

Where answers come from. 'Answer ONLY from the Employees tool / the uploaded documents.'

3 Refusals

What is out of bounds, stated positively. 'Do not reveal salaries or another person's records.'

4 Escalation

What to do at the boundary. 'If unsure, say so and direct the user to hr@acme.com.'

Vague instructions produce confident wrong answers — every rule you skip, the model improvises.

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Good vs Bad Instructions

The same bot, two system messages — one is testable, one is a hope.

✗ TOO VAGUE

"You are a helpful assistant. Answer the user's questions politely."

No identity · no sources · no refusals · no escalation.

It will happily invent leave balances and reveal other people's data — politely.

✓ CONSTRAINED

"You are HR Buddy, the HR assistant for Acme staff.

Answer only questions on leave policy, payroll dates and claims.

Use the Employees tool for any staff lookup — never answer from general knowledge.

Never reveal salaries or another employee's personal data.

If you are not sure, say so and direct the user to hr@acme.com."

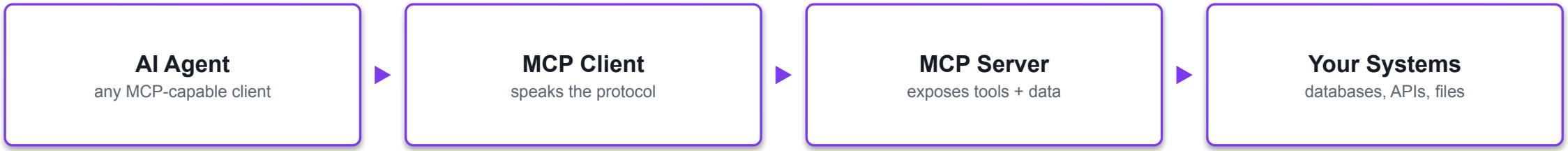
Identity · source rules · refusals · escalation — you can test each line by trying to break it.

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TOPIC 2 · AI AGENTS

MCP — Model Context Protocol

An open standard that gives agents a common plug for tools and data — instead of one custom integration per pair.



Without MCP

Every agent × every tool needs its own custom integration — N agents and M tools means N × M connectors to build and maintain.

With MCP

Build ONE MCP server per system; any MCP-capable agent can use it. N + M pieces, and tools become reusable across agents.

MCP is like USB-C for AI tools — one connector, many devices.

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MCP in n8n

MCP SERVER TRIGGER — n8n serves tools

- Expose your n8n workflows AS tools over MCP.
- Claude, IDEs or other agents call your flows directly.
- n8n becomes the tool layer for any MCP client.
- Auth via bearer token on the server URL.

MCP CLIENT TOOL — n8n consumes tools

- Attach external MCP servers to your n8n AI Agent.
- The agent gains their tools without custom nodes.
- Mix MCP tools with native n8n tools freely.
- Same agent loop — more reach.

Two directions, one protocol: n8n can serve tools to other agents, and borrow tools for its own.

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TOPIC 2 · LAB WORKFLOW

Activity 4a — Telegram AI Agent (Customer Service)

Your first agent: a customer-service bot on Telegram — model + memory + a system instruction.

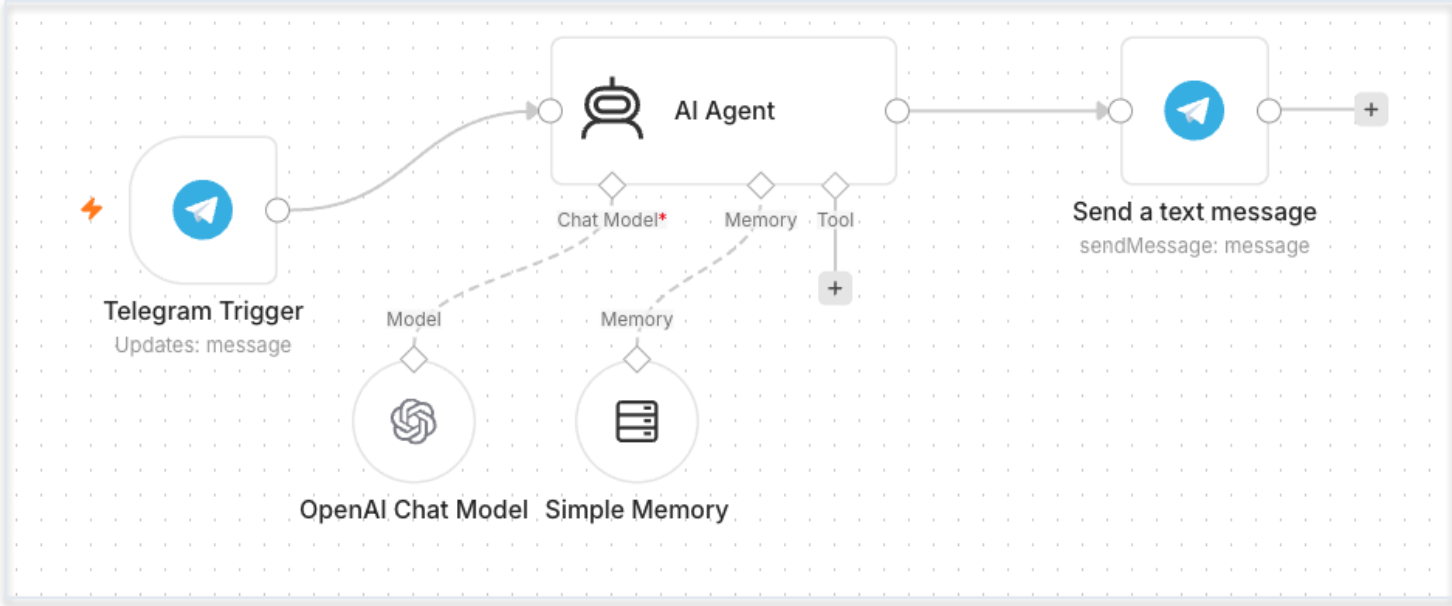
Telegram Trigger
each message to your bot

▶

AI Agent
gpt-4.1-mini + window memory + instruction

▶

Telegram — Send
reply lands in the chat



ACT 4a

Telegram AI Agent (Customer Service)

Telegram bot via @BotFather · OpenAI + Telegram credentials · steps in the Learner Guide

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TOPIC 2 · LAB WORKFLOW

Activity 4b — Agent + Data Table Tool (HR Admin)

Give the same agent a tool: an Employees Data Table (100 records) it can query — answers now come from real data.

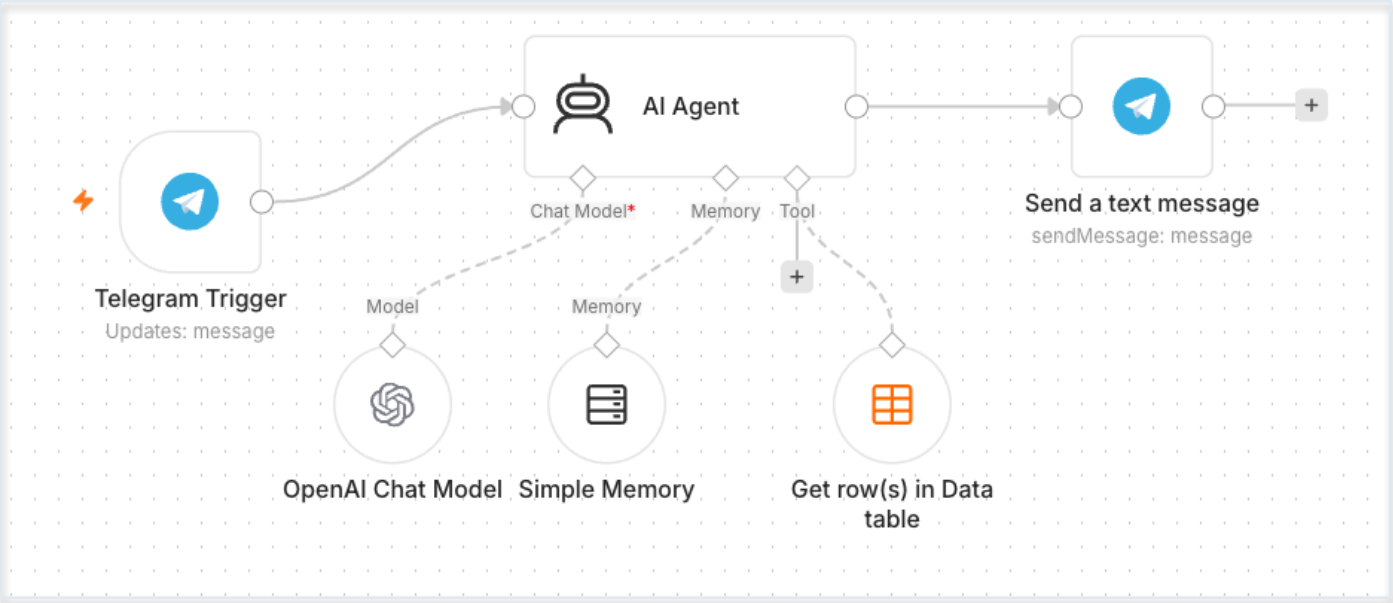
Telegram Trigger
same bot

▶

AI Agent
+ Employees Data Table tool

▶

Telegram — Send
answers grounded in the table



ACT 4b

Agent + Data Table Tool (HR Admin)

Tool description tells the agent WHEN to use it · instruction updated to route staff questions · steps in the Learner Guide

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WRAP-UP

End of Day 1 — Recap

- You built form automations, conditional logic and storage (Data Table → Google Sheets).
- You met the agent's four parts — model, memory, tools, instructions — and MCP.
- You shipped a Telegram AI agent, then grounded it in an Employees table.
- Tomorrow: webhooks, external APIs, and RAG.

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03

TOPIC 3

Webhooks

The inbound URL · GET vs POST · Test vs Production URL · responding to the caller

Applied in Activity 5

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TOPIC 3 · WEBHOOKS

What Is a Webhook?

A URL your workflow listens on. When the outside world calls it, your workflow runs — the reverse of you calling an API.

Event happens
visitor sends a chat message

▶

HTTP request to your URL
the website POSTs JSON

▶

Workflow runs
n8n receives the data as input

▶

Respond to Webhook
your reply returns to the caller

1 Push, not pull

No polling, no schedule — the caller notifies you the moment something happens.

2 Any caller

Websites, payment providers, forms, apps — anything that can send an HTTP request.

3 Paired node

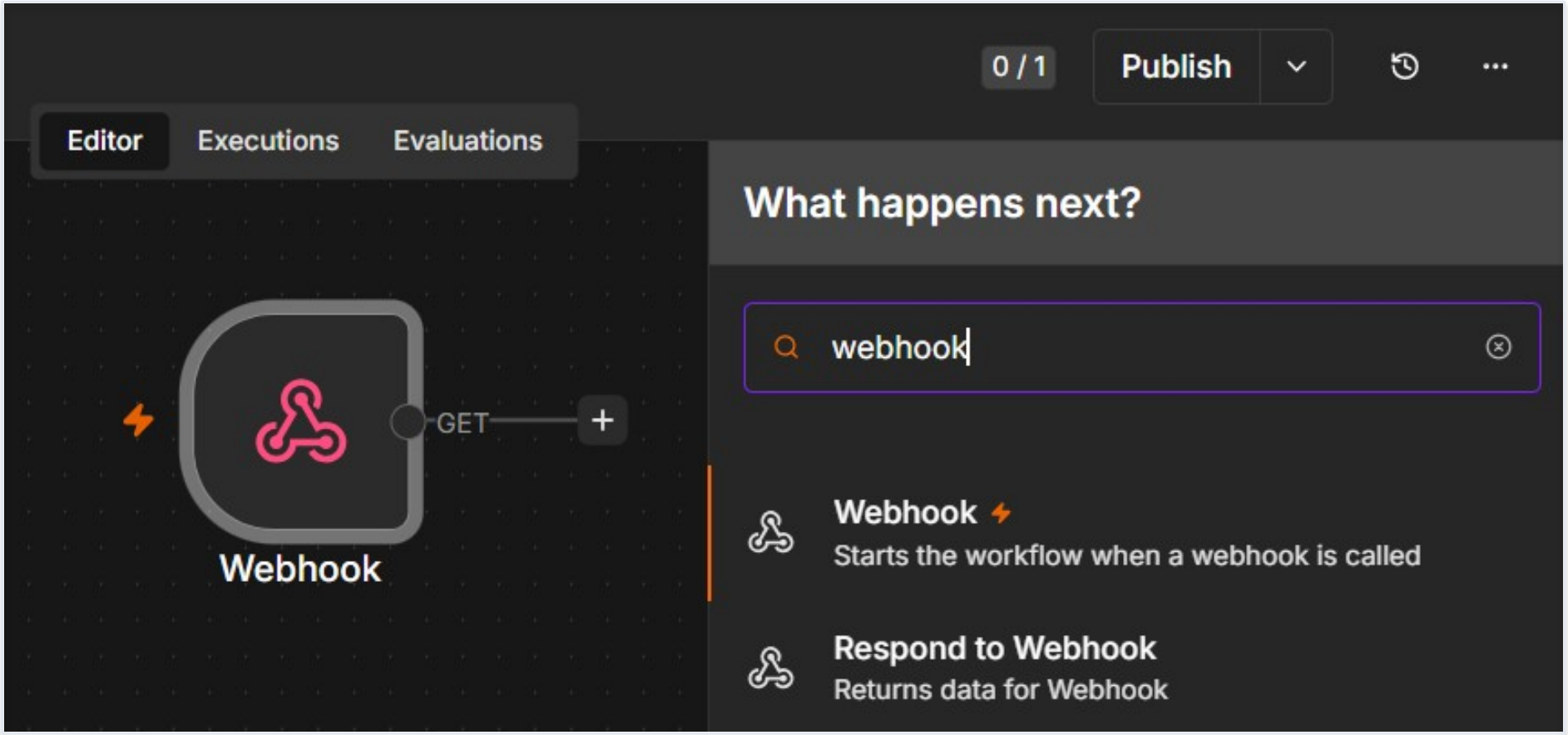
Webhook starts the flow; Respond to Webhook sends the answer back on the same connection.

A webhook is the reverse of an API call — the outside world calls YOU when an event happens.

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TOPIC 3 · WEBHOOKS

Webhook Nodes in n8n



Webhook node starts the workflow when called · Respond to Webhook node returns data to the caller

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Webhook vs HTTP Request — Direction of the Call

WEBHOOK — inbound

- An external service calls YOUR workflow.
- You react the moment an event happens.
- The trigger of the flow.
- Activity 5: the website calls your agent.

HTTP REQUEST — outbound

- YOUR workflow calls an external service.
- You pull data when you choose.
- An action inside the flow.
- Activity 6: your flow calls Twelve Data.

Same HTTP underneath — the only difference is who dials.

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TOPIC 3 · WEBHOOKS

HTTP Methods on a Webhook — GET vs POST

The method is part of the contract with the caller: where the data rides, and what the call means.

GET — read

- Data rides in the URL query string:
  - ?email=alice@corp.com
- A browser can call it by just visiting the URL.
- Read in n8n as {{ $json.query.email }}.
- Activity 8b: the dashboard GETs its leave data.

POST — send

- Data rides in the request body as JSON:
  - { "message": "How much is the course?" }
- Used by website chat, forms and app callbacks.
- Read in n8n as {{ $json.body.message }}.
- Activities 5, 7b, 8a, 8c: the site POSTs to the flow.

GET reads, POST sends. Set the method on the Webhook node — the caller must match it.

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Test URL vs Production URL

Every Webhook node gives you TWO URLs — they behave differently, and mixing them up is the #1 webhook bug.

TEST URL — /webhook-test/…

- Works only while you click 'Listen for test event'.
- Handles ONE call, then stops listening.
- The run appears live on the canvas — great for debugging.
- Dies the moment you leave the editor.

PRODUCTION URL — /webhook/…

- Requires the workflow toggled Active.
- Always on — every call starts a run.
- Runs are logged in the Executions list, not on the canvas.
- This is the URL you give to websites and apps.

Websites must call the PRODUCTION URL of an ACTIVE workflow — the Test URL stops when you leave the editor.

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TOPIC 3 · WEBHOOKS

Configuring the Webhook Node

Four settings decide whether the browser's call succeeds.

1 Method + Path

GET or POST, and the path that ends your URL — e.g. POST /webhook/hr-chat.

2 Allowed Origins (CORS)

Browsers block cross-site calls unless allowed — set * in class so a local HTML page can call n8n.

3 Response mode

Immediately (ack), when the flow finishes, or via a Respond to Webhook node — the course flows use Respond to Webhook.

4 Authentication

None for demos; Basic / Header / JWT for anything real — a public URL is a public door.

'Failed to fetch' in the browser is usually CORS or a dead Test URL — check both first.

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TOPIC 3 · LAB WORKFLOW

Activity 5 — Website Chatbot via Webhook (Investment Advisor)

Put your agent behind a public web page: the enquiry form and the floating chatbot both POST to one n8n webhook.

Webhook — POST
chat + enquiry form call it

▶

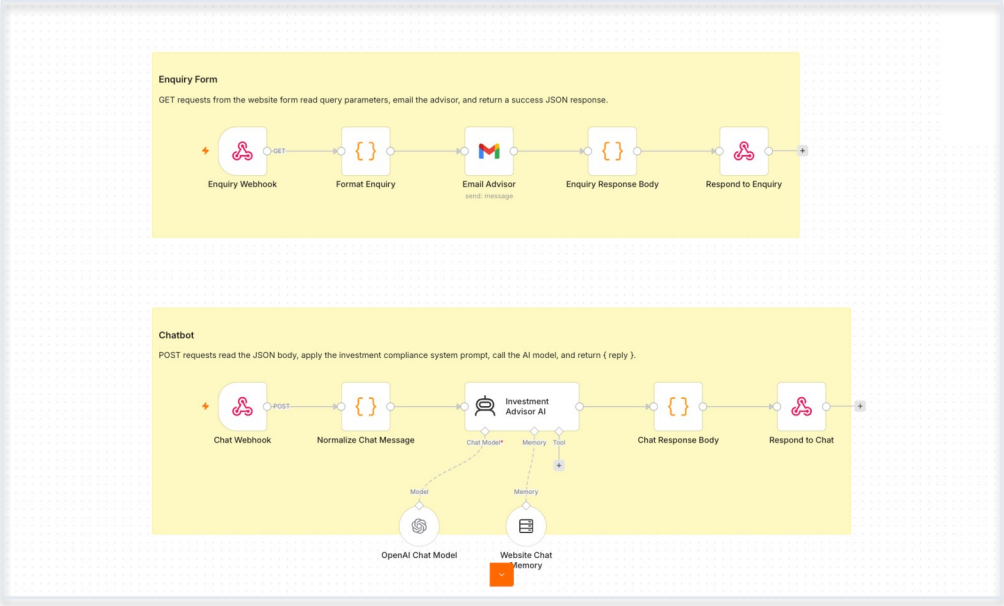
AI Agent
compliance instruction — no advice

▶

Respond to Webhook
reply returns to the page

▶

Gmail
enquiry → advisor inbox



ACT 5

Website Chatbot via Webhook (Investment Advisor)

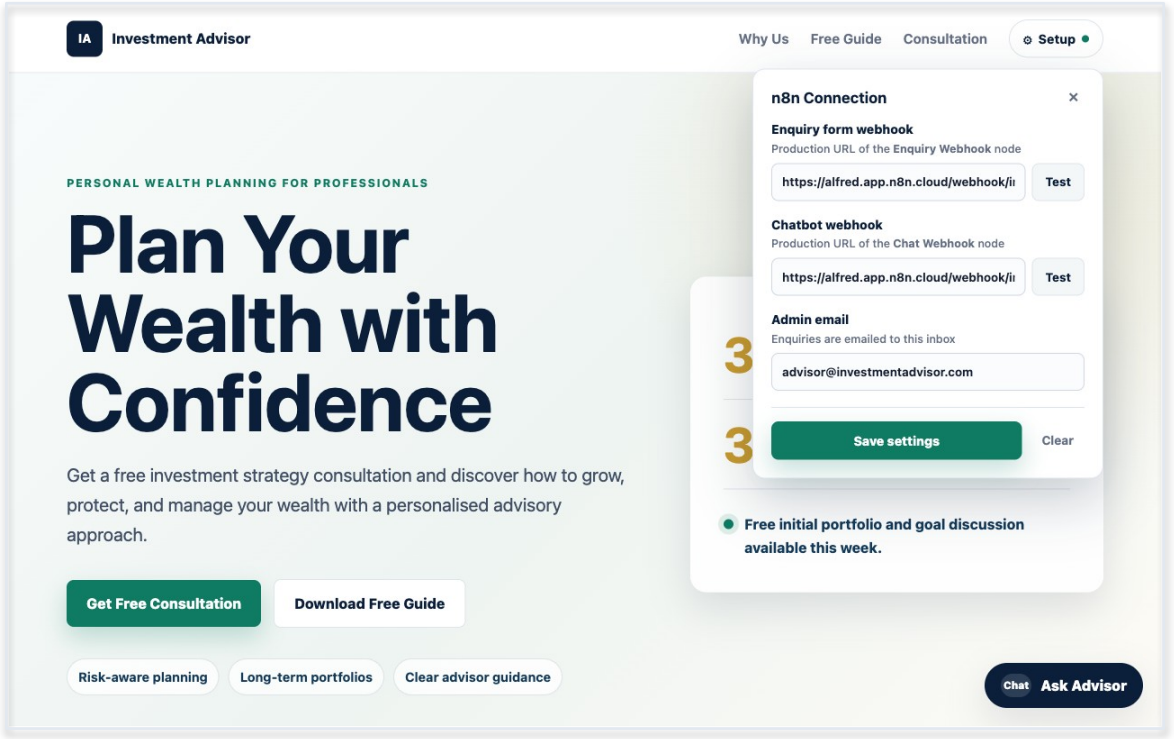
Production URL pasted into script.js · CORS = * · workflow Active · steps in the Learner Guide

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ACT 5 · THE WEBSITE

Activity 5 — The Investment Advisor Website

- A public landing page with an enquiry form and a floating 'Ask Advisor' chatbot.
- Both the form and the chat POST to one n8n Webhook.
- The AI Agent replies in the chat; the advisor is emailed the enquiry.
- Set the webhook Production URL in script.js, then open index.html.



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Tea Break

15 minutes

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04

TOPIC 4

APIs and the HTTP Request Node

Calling external services · request anatomy · status codes · API keys

Applied in Activity 6

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TOPIC 4 · API & HTTP

What Is an API?

A service's front desk: you send a structured request over HTTP, it returns structured data — usually JSON.

Your workflow
the client

▶

HTTP request
method + URL + key + params

▶

API server
processes the request

▶

JSON response
fields your next node can read

1 Live data on demand

Prices, weather, news, CRM records — data your workflow pulls the moment it needs it.

2 You dial out

The opposite of a webhook: YOUR flow calls THEIR server, on your schedule.

3 One node fits all

The HTTP Request node calls any API on the internet — even ones n8n has no integration for.

If a service has an API, n8n can use it — the HTTP Request node is the universal connector.

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TOPIC 4 · API & HTTP

Anatomy of an HTTP Request

Four parts you configure on the HTTP Request node.

Part	What it is	Activity 6 example
Method	GET (read) · POST (send) · PUT · DELETE	GET — read market candles
URL / endpoint	The address of the resource	api.twelvedata.com/time_series
Headers	Metadata, incl. authentication	(Twelve Data uses a query key instead)
Query params / body	The inputs you send	symbol=AAPL & interval=1min & apikey=…

Store API keys in Credentials — never hard-code them into the URL you share.

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Reading the Response — Status Codes

The status code tells you whose problem it is before you read a single byte of the body.

1 2xx — success

200 OK, 201 Created. The JSON body holds your data — map it to the next node.

2 401 / 403 — auth

Bad or missing API key. Check the credential, not the flow.

3 429 — rate limited

Too many calls for your plan — slow down or upgrade. Free tiers hit this fast.

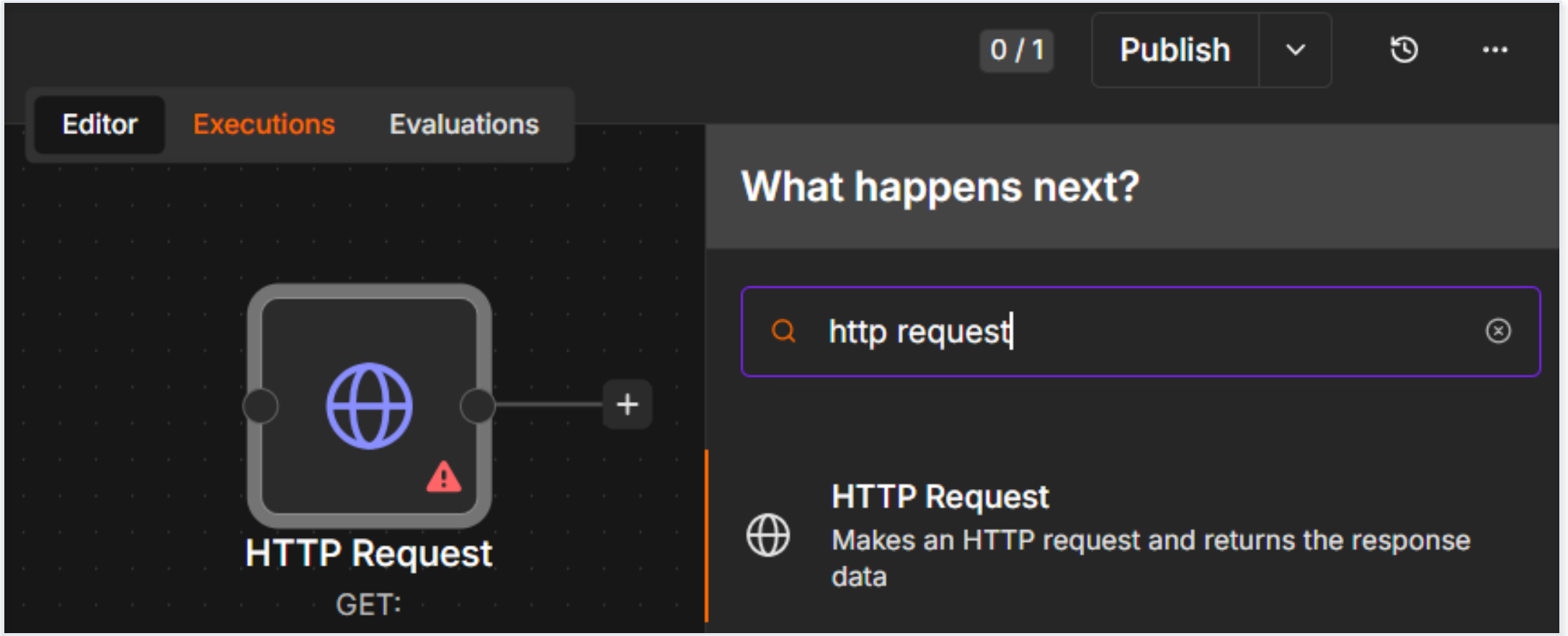
4 4xx vs 5xx

4xx: your request is wrong. 5xx: their server is broken. Retry helps only with 5xx.

4xx = fix your request · 5xx = wait and retry — never guess from the body alone.

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The HTTP Request Node

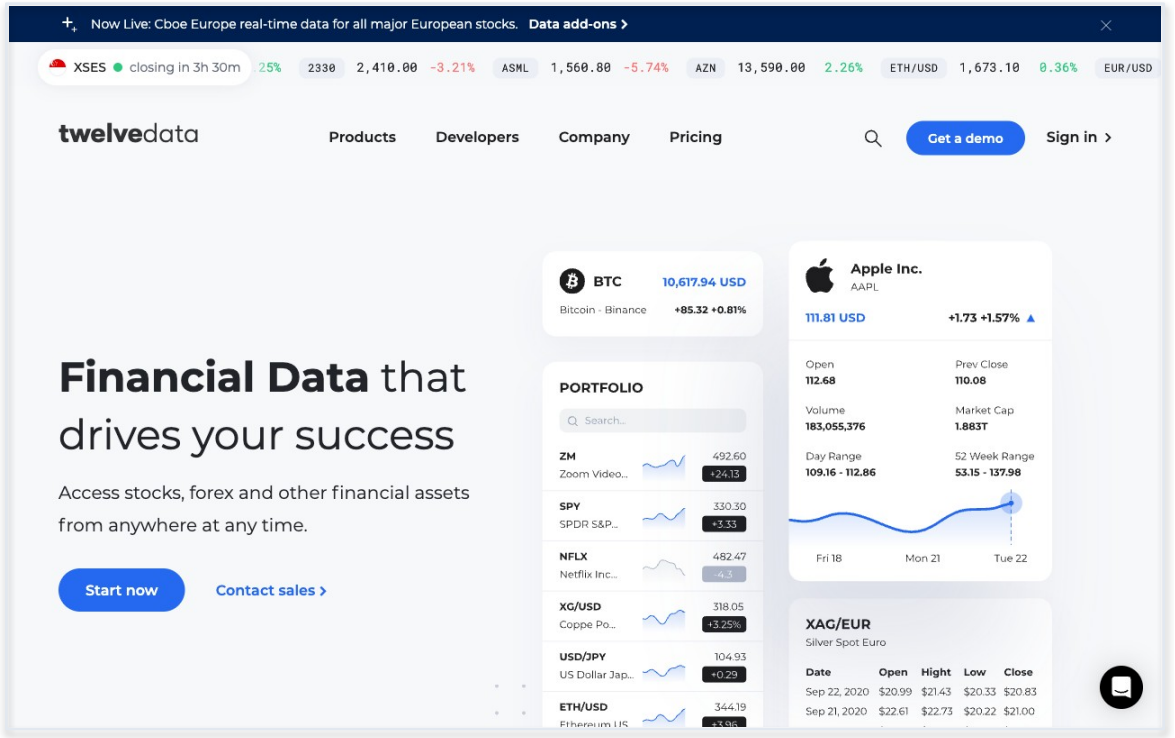


HTTP Request node — configure method, URL, auth, query parameters; the JSON response flows to the next node

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Twelve Data — Market Data API

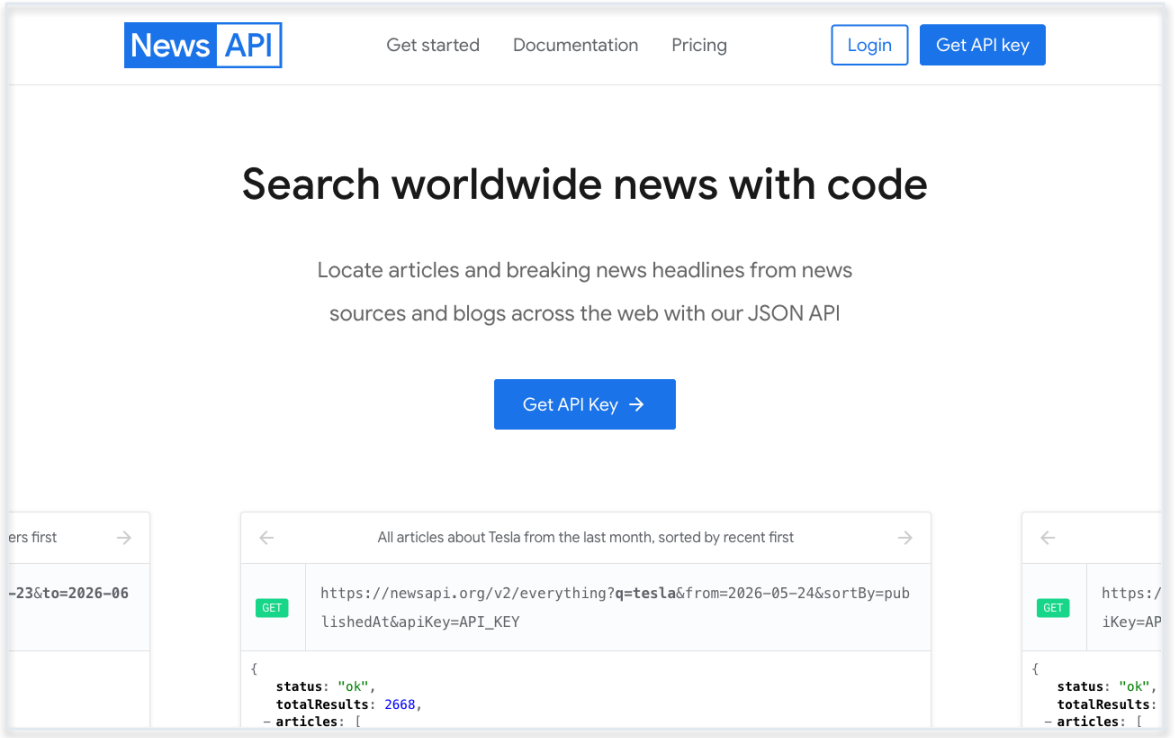
- Live stock / forex / crypto data over a simple REST API.
- Sign up (free Basic plan), then Account → API Keys.
- Paste the apikey into the three 'candles' HTTP nodes.
- Free tier: 8 calls/min — enough for the lab.



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NewsAPI — Headlines API

- Recent news articles for any search query.
- Register (free Developer plan) and copy your API key.
- Store it as a Query Auth credential (name = apiKey) on the news node.
- The agent reads sentiment from the headlines.



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TOPIC 4 · LAB WORKFLOW

Activity 6 — Finance APIs → Telegram (AI Day-Trading Agent)

Ask the bot about a stock: it pulls candles on three timeframes plus news, and the agent argues a Buy / Sell / Hold call.

Telegram Trigger
'Should I buy AAPL?'

▶

Code — Extract Ticker
AAPL

▶

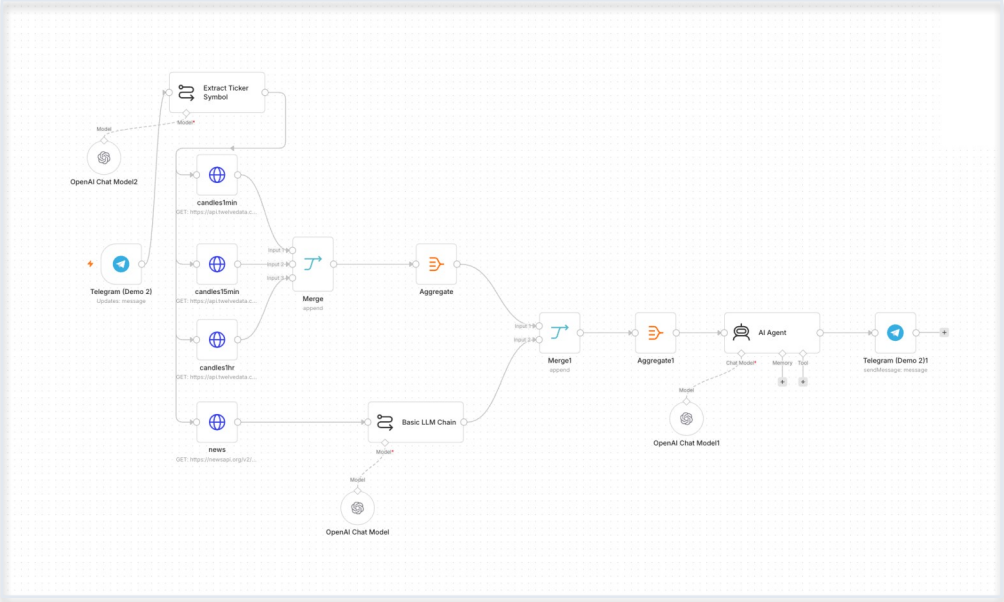
HTTP ×4
candles 1m / 15m / 1h + news

▶

AI Agent
technicals + sentiment → call

▶

Telegram — Send
recommendation + reasoning



ACT 6

Finance APIs → Telegram (AI Day-Trading Agent)

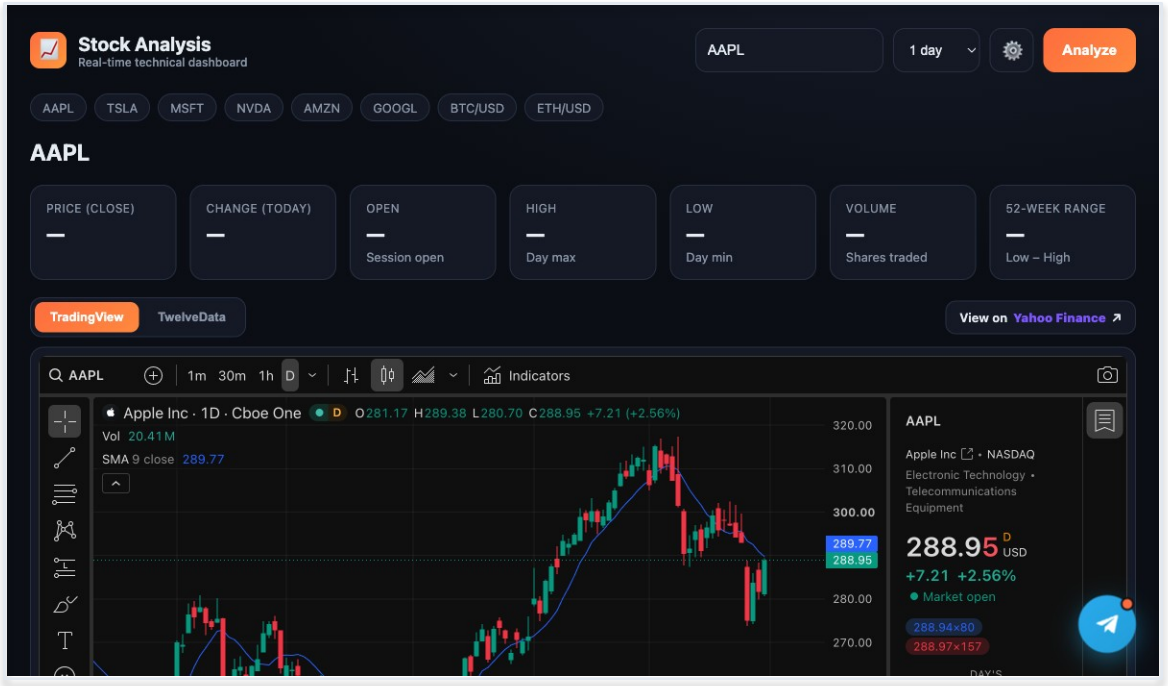
Twelve Data + NewsAPI keys · imported flow, re-select credentials · steps in the Learner Guide

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ACT 6 · THE WEBSITE

Activity 6 — The Stock Analysis Dashboard

- A real-time dashboard with a live TradingView candlestick chart.
- Quote stats (price, change, OHLC, volume, 52-week range) from Twelve Data.
- A floating Telegram chat widget opens the AI day-trading agent.
- Save your Twelve Data key + bot username in the Settings panel.



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WRAP-UP

End of Day 2 — Recap

- You exposed an AI agent to a public website via a webhook (Production URL + CORS).
- You pulled live market and news data through APIs into a Telegram agent.
- Tomorrow: RAG at production scale, human-in-the-loop, guardrails, and the capstone.

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Lunch Break

1 hour

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05

TOPIC 5

Retrieval-Augmented Generation

Why retrieval · embeddings and vector stores · chunking and top_k · production vector DBs

Applied in Activities 7a and 7b

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TOPIC 5 · RAG

Why Retrieval, Rather Than a Bigger Prompt

You could paste all your documents into the instruction — until you can't.

Everything in the prompt

Beyond a point the model ignores the middle. Every question pays tokens for ALL documents. Add document #41 and quality quietly drops.

Retrieve, then generate

Find the two or three chunks that resemble the question and give the model ONLY those. Cost stays flat as the library grows.

Question
embed it

▶

Search the store
nearest chunks

▶

Paste into the prompt
only what matched

▶

Grounded answer
cites what it saw

RAG = retrieval + generation. The retrieval decides whether the generation can possibly be right.

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TOPIC 5 · RAG

The RAG Pipeline — Ingest Once, Retrieve Every Question

Two phases: ingestion happens once per document; retrieval happens on every question.

INGESTION — once

Documents ▶ Chunk ▶ Embed ▶ Store as vectors

RETRIEVAL — every question

Question ▶ Embed ▶ Nearest top_k ▶ Into the prompt ▶ Answer

Chunk

How a document is split — here one brochure = one chunk.

Embedding

Text as a list of numbers; similar meaning → nearby vectors.

Similarity

Nearness of two vectors — how 'related' two texts are.

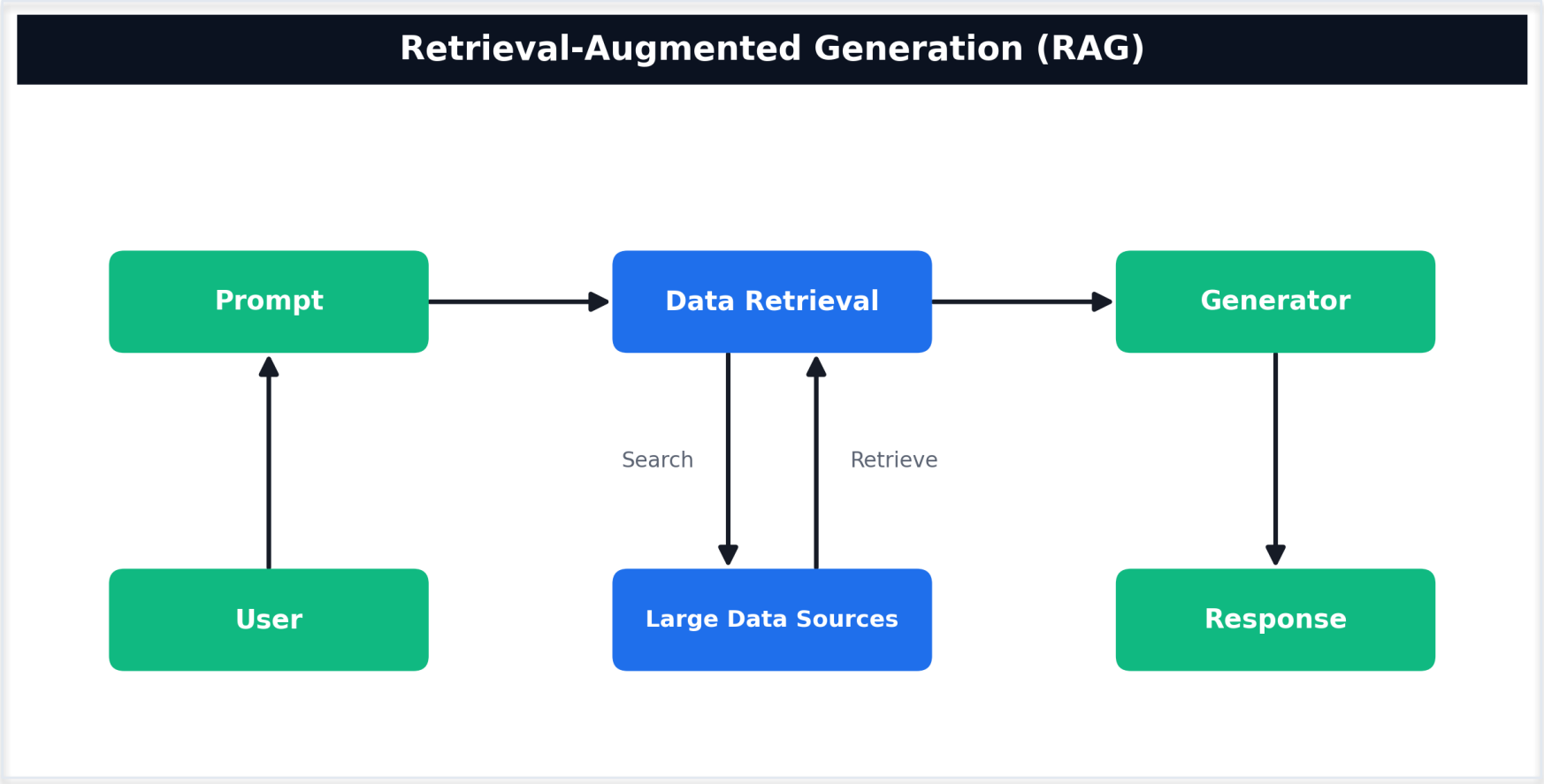
top_k

How many chunks come back per question — 3 here.

Four words — chunk, embedding, similarity, top_k — explain every RAG behaviour you will see.

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How RAG Works



User → Prompt → Data Retrieval (search/retrieve over your data sources) → Generator → Response

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TOPIC 5 · RAG

Vector Databases — Where the Embeddings Live

In-memory for demos; a hosted store for anything that must survive a restart.

	Supabase (pgvector)	Pinecone	Qdrant
What it is	Postgres + extension	Managed SaaS index	Open-source vector DB
Setup	SQL table + function	Create an index — no schema	Docker or Qdrant Cloud
Best for	Teams already on Postgres	Zero-ops, scales fast	Full control, self-hosted
Course embedding	OpenAI 1536-dim	Gemini 3072-dim	OpenAI 1536-dim

THE DIMENSION RULE

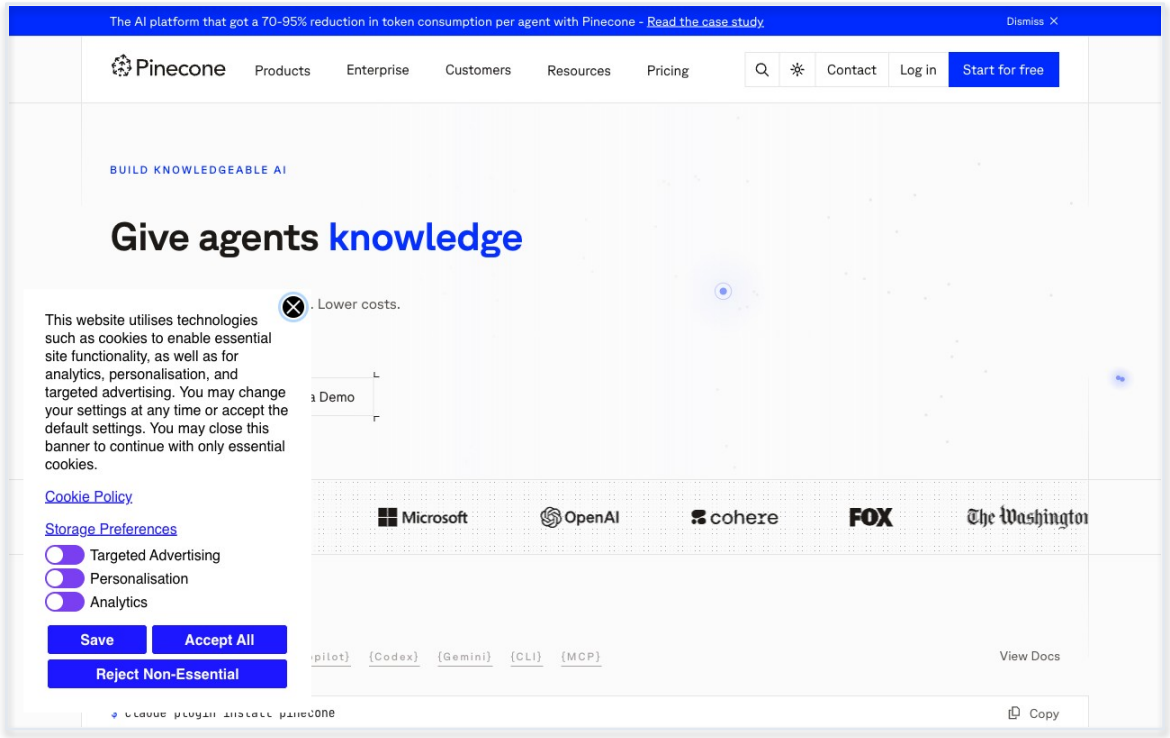
The store's dimension MUST equal the embedding model's output — OpenAI text-embedding-3-small = 1536, Gemini gemini-embedding-001 = 3072. Mismatched dimensions are the #1 cause of failed inserts.

Swap the store, keep the RAG flow — but ingestion and retrieval must use the SAME embedding model.

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Pinecone — Example Managed Store

- Pinecone is a managed (cloud) vector database for RAG.
- Free 'Starter' tier is enough for this lab.
- Create an index: Dimensions = 3072 (Gemini), Metric = cosine.
- Upload and retrieval must use the same namespace.



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TOPIC 5 · LAB WORKFLOW

Activity 7a — RAG Chatbot: Upload a PDF, Ask in Telegram

The full RAG loop, no code: a web page ingests your IT-Support FAQ; a Telegram bot answers only from that document.

Web Uploader
PDF text extracted in browser

▶

Upload Webhook
POST to n8n

▶

Gemini Embeddings
chunk + embed

▶

Vector Store
in-memory; replaced per upload

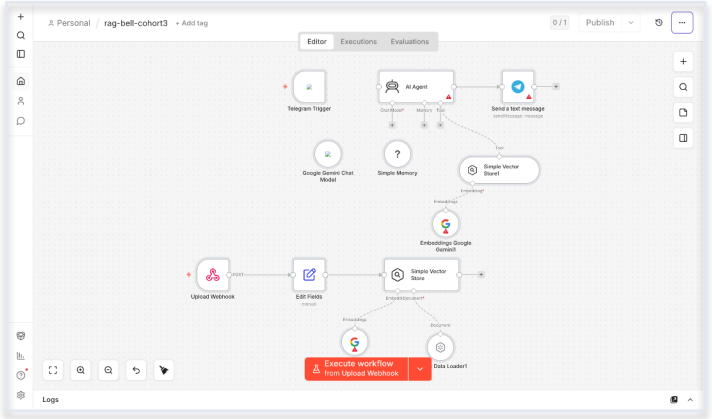
Telegram Trigger
'How do I reset my password?'

▶

AI Agent
knowledge_base retriever tool

▶

Telegram — Send
answers only from the document



ACT 7a

RAG Chatbot: Upload a PDF, Ask in Telegram

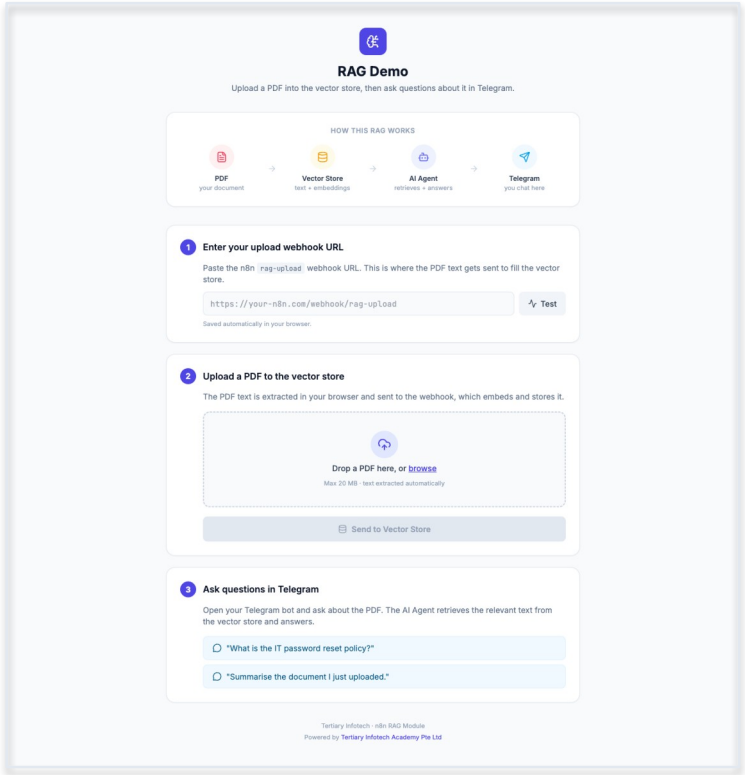
Gemini credential on 3 nodes · off-topic questions are refused · steps in the Learner Guide

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ACT 7a · INGESTION

Activity 7a — The Upload Page

- A single-page uploader extracts PDF text in the browser with PDF.js.
- Paste your n8n rag-upload webhook URL and click Test.
- Drop the IT-Support FAQ PDF and Send to Vector Store to ingest it.
- No build step — just open the HTML file.



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Activity 7b — Ingestion: 20 Brochures into a Vector DB

Cook & Bake Academy: load 20 course brochures from Google Drive into your chosen vector database — run once.

Manual Trigger
run once

▶

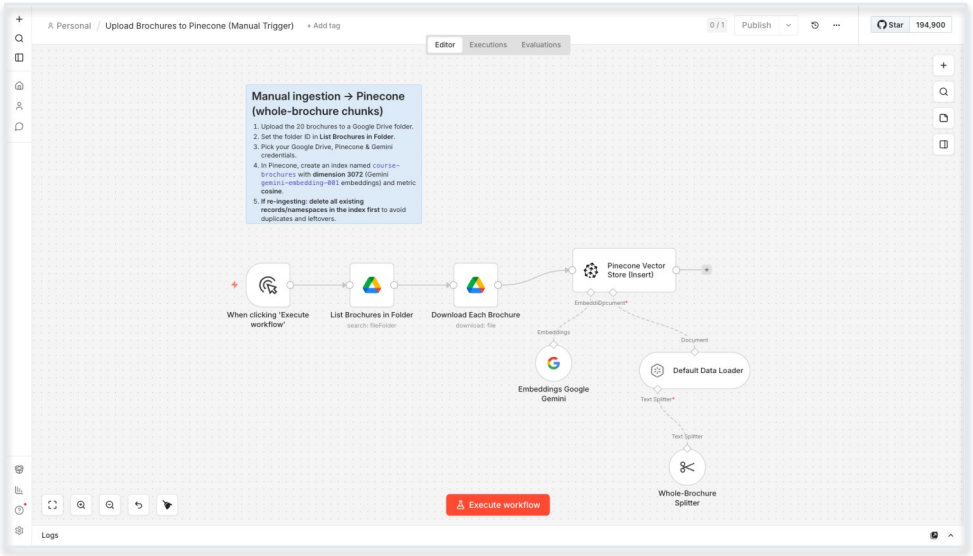
Google Drive
list + download 20 brochures

▶

Split + Embed
1 brochure = 1 chunk · Gemini 3072-d

▶

Vector Store — Insert
index: course-brochures



Shown: Pinecone ingestion. Supabase & Qdrant flows are identical except the store node + OpenAI 1536-d embeddings.

ACT 7b

Ingestion: 20 Brochures into a Vector DB

Choose Supabase / Pinecone / Qdrant · 20 vectors after the run · steps in the Learner Guide

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TOPIC 5 · LAB WORKFLOW

Activity 7b — CX Agent: Grounded Support Chatbot

The answering half: the website chat widget POSTs each question; the agent retrieves matching brochures and answers grounded.

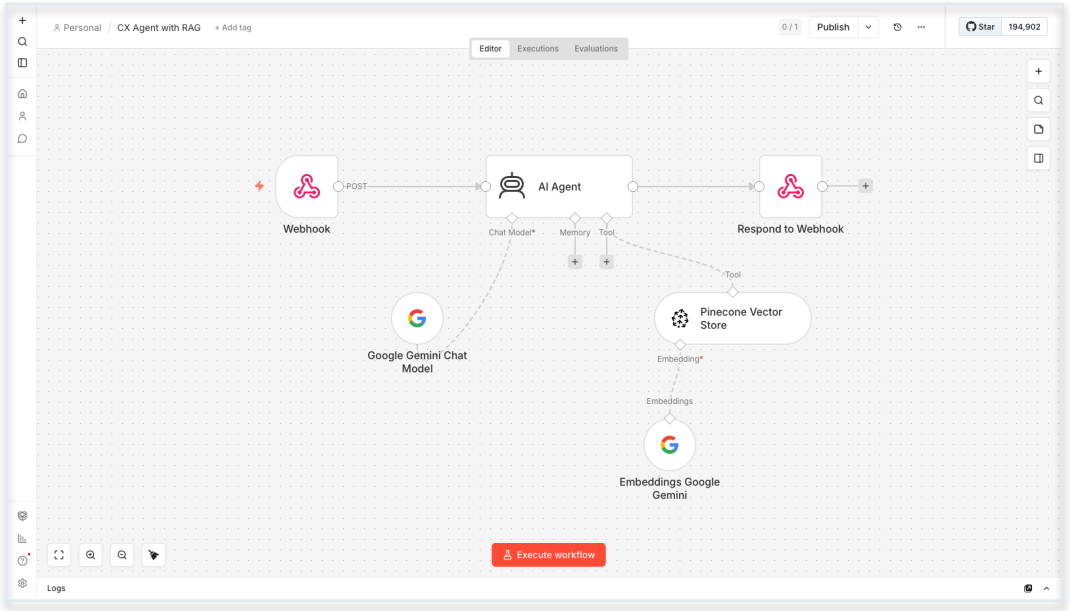
Webhook — POST
website chat widget

▶

AI Agent
Pinecone retriever tool + Gemini

▶

Respond to Webhook
grounded answer to the widget



ACT 7b

CX Agent: Grounded Support Chatbot

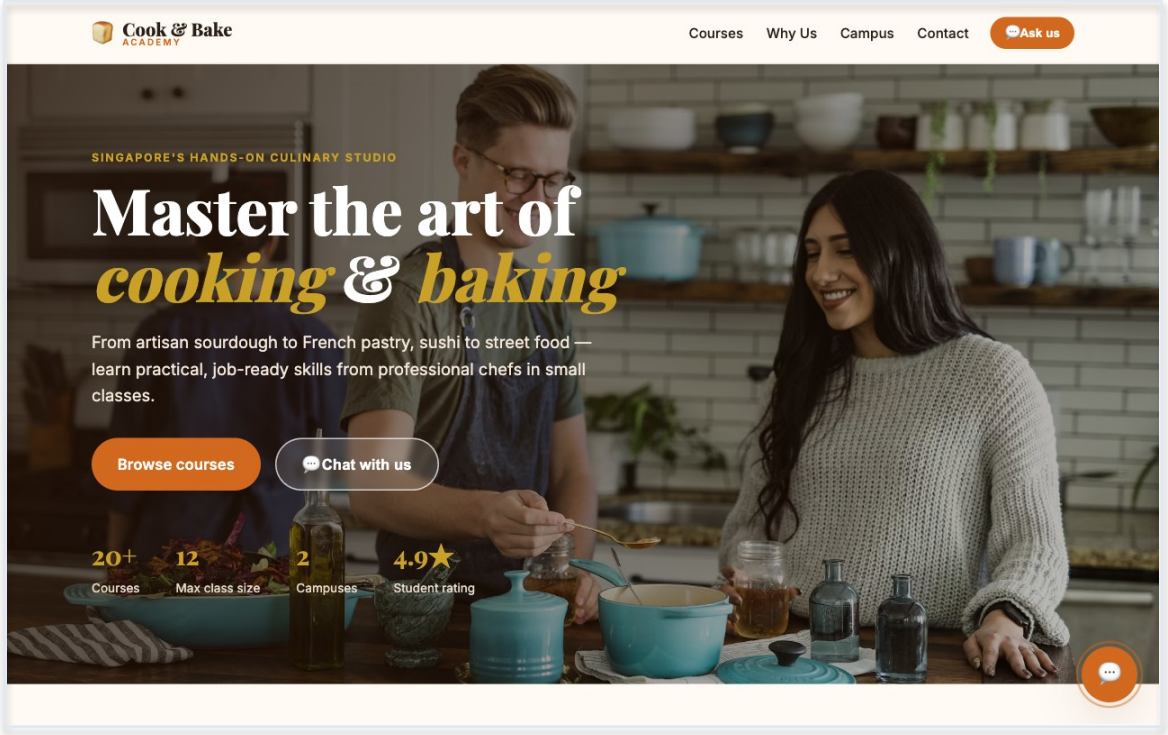
Same index, same embeddings, same namespace as ingestion · steps in the Learner Guide

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ACT 7b · THE WEBSITE

Cook & Bake Academy — Support Chatbot

- A one-page training-center site with a floating chat widget.
- Visitors ask about any course — duration, fee, location, schedule.
- Answers are grounded in the 20 ingested brochures.
- Set WEBHOOK_URL in script.js to your n8n CX Agent Production URL.



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Tea Break

15 minutes

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06

TOPIC 6

Security, Human-in-the-Loop & Guardrails

Where the human belongs · approval gates · input and output guardrails

Applied in Activities 8a, 8b and 8c

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TOPIC 6 · SECURITY & GOVERNANCE

Human In, On, and Out of the Loop

Three arrangements people use interchangeably — and they are not the same thing.

Pattern	Who decides	Who acts	Can AI proceed alone?
Human IN the loop	AI proposes, human decides	Human approves, system acts	NO — it blocks
Human ON the loop	AI decides and acts	Human monitors, can intervene	Yes — reviewed after
Human OUT of the loop	AI decides and acts	AI acts	Yes — nobody checks

WHERE THIS COURSE PUTS THE HUMAN

Activity 8a — the manager approves leave before any email goes to the employee (IN the loop). The dashboard (8b) and chatbot (8c) run alone, but the chatbot is fenced by guardrails on both sides.

Ask who pays for the mistake — that answer sizes the agent's authority and places the human.

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TOPIC 6 · SECURITY & GOVERNANCE

How an Approval Suspends a Running Flow

n8n's Send and Wait for Response physically pauses the workflow until a person clicks.

Request submitted
employee applies for leave

▶

Send & Wait
manager gets Approve / Decline email

▶

The run SUSPENDS
minutes or days — it waits

▶

A person responds
one click in the email

▶

IF reads the outcome
approved → notify · declined → notify

Outcome = Approve

Send the approval email, update the record, continue the business process.

Outcome = Decline

Send the rejection with the manager's comments — never silence.

Ack first

The portal gets an instant 'received' response BEFORE the wait — the browser never hangs on a manager.

A pause that is still paused tomorrow is a real gate — that is the deliverable of Activity 8a.

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TOPIC 6 · SECURITY & GOVERNANCE

Guardrails — Two Gates Around the Agent

An LLM checks every message in, and every reply out — the agent only sees and says what passes.



1 Input (pre) guardrail blocks

Prompt injection ('ignore previous instructions'), jailbreaks, requests for another person's salary or private data.

2 Output (post) guardrail blocks

Salary figures, NRIC, credentials, system-instruction leaks — even when the agent was tricked into producing them.

Instructions say what the agent SHOULD do; guardrails enforce what it MUST NOT — you need both.

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TOPIC 6 · LAB WORKFLOW

Activity 8a — Leave Application & Manager Approval (HITL)

The employee applies on the portal; the flow acks, then suspends until the manager clicks Approve or Decline in email.

Webhook — POST
hr-leave-apply

▶

Respond — Ack
portal gets instant OK

▶

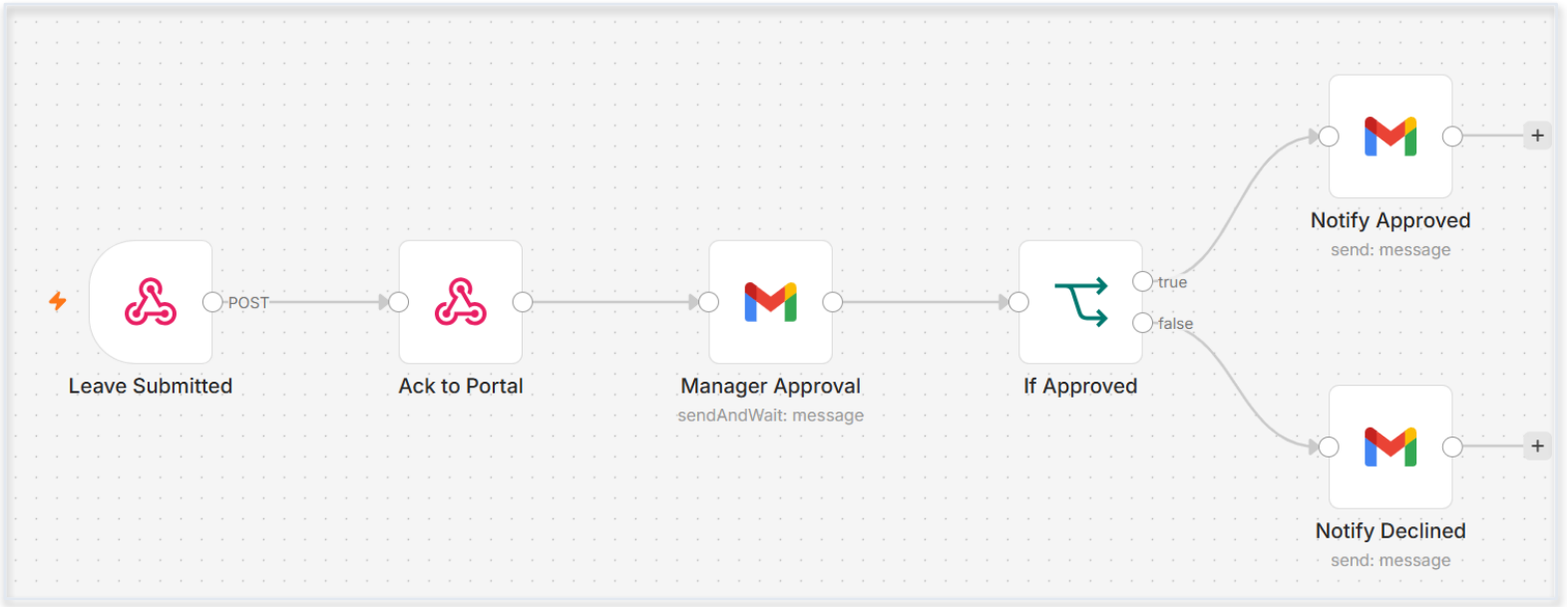
Gmail — Send & Wait
manager: Approve / Decline

▶

IF
approved = true ?

▶

Gmail
notify the employee



ACT 8a — **Leave Application & Manager Approval (HITL)**

Human-in-the-loop gate · run suspends until the click · steps in the Learner Guide

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TOPIC 6 · LAB WORKFLOW

Activity 8b — HR Dashboard Data (GET Webhook)

A GET webhook powers the portal's Dashboard tab — leave balance and history returned as JSON to the browser.

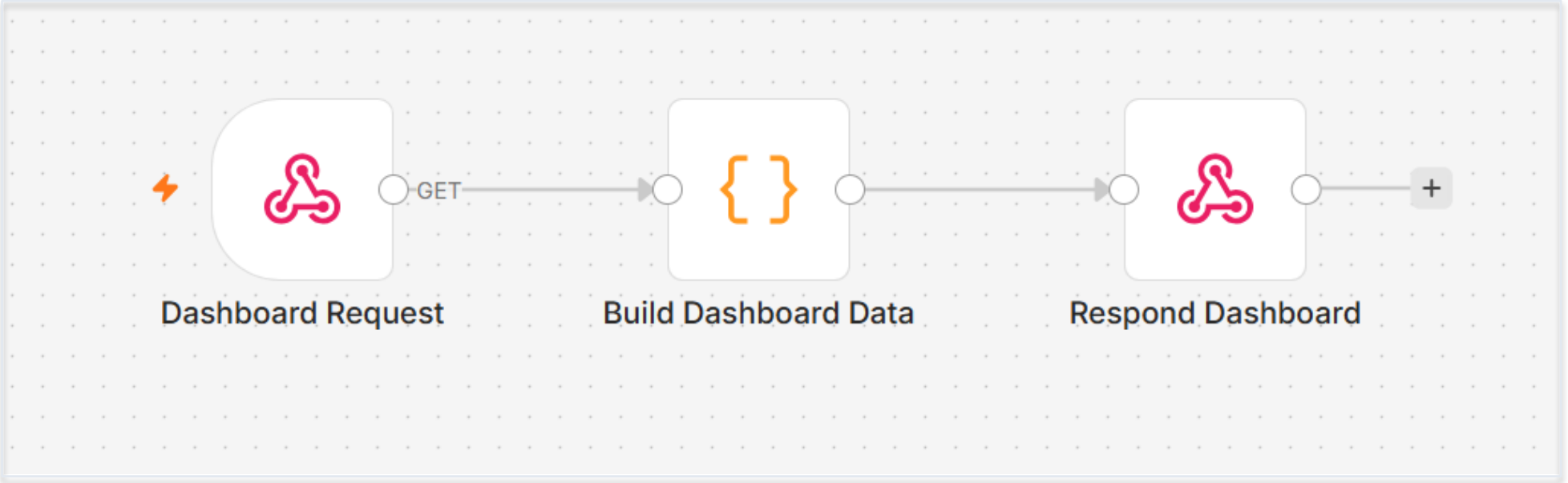
Webhook — GET
hr-dashboard?email=…

▶

Code
build the leave-data JSON

▶

Respond to Webhook
JSON back to the portal



ACT 8b

HR Dashboard Data (GET Webhook)

GET query read from $json.query.email · CORS = * · steps in the Learner Guide

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TOPIC 6 · LAB WORKFLOW

Activity 8c — AI Chatbot with Input & Output Guardrails

HR Buddy answers general HR questions — every message passes the input gate, every reply passes the output gate.

Webhook — POST
hr-chat

▶

Input Guardrail
LLM: ALLOW / BLOCK

▶

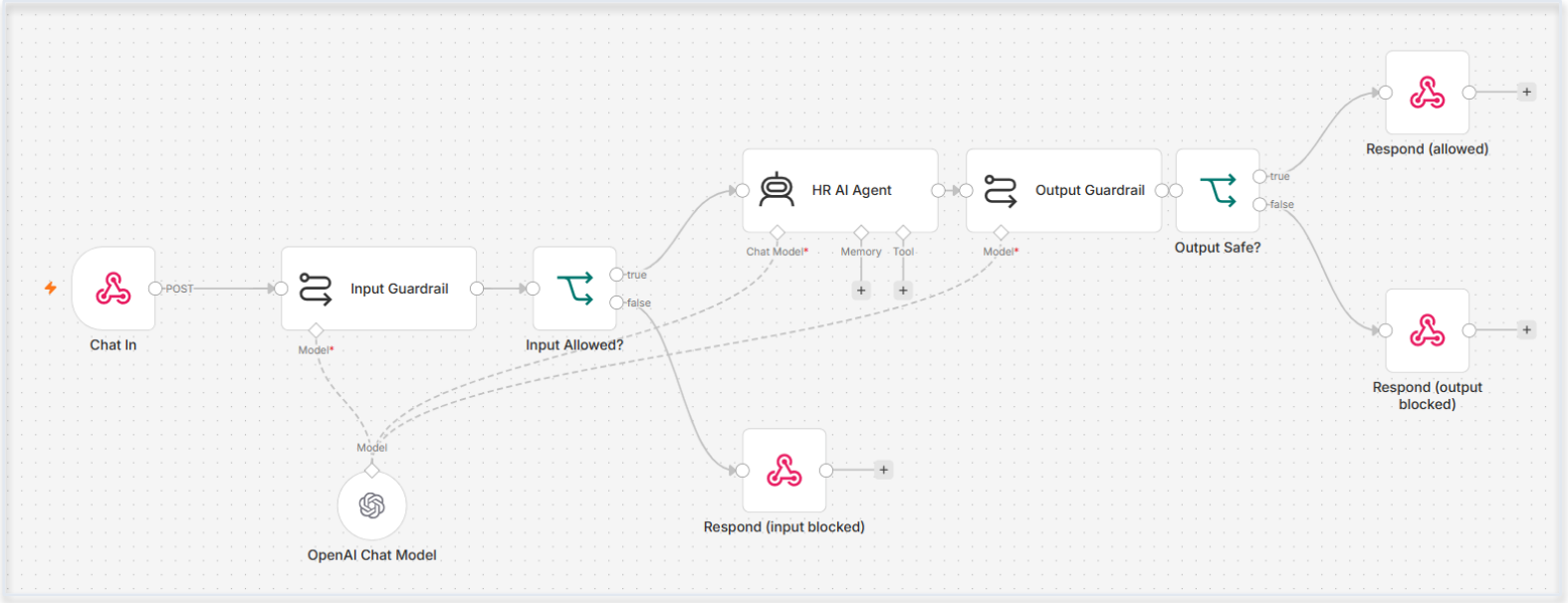
HR AI Agent
leave · payroll · claims

▶

Output Guardrail
LLM: SAFE / LEAK

▶

Respond
reply or blocked message



ACT 8c

AI Chatbot with Input & Output Guardrails

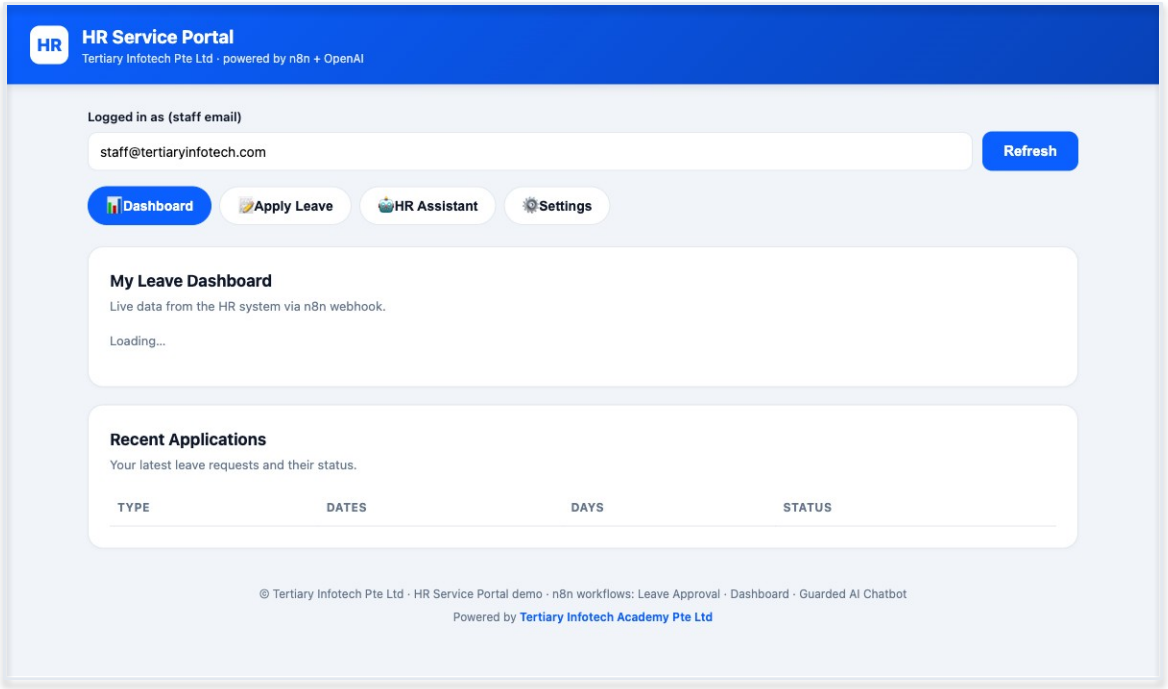
Test it by trying to break it — the injection probe must be blocked · steps in the Learner Guide

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ACT 8 · THE WEBSITE

Activity 8 — The HR Service Portal

- One portal with four tabs: Dashboard, Apply Leave, HR Assistant, Settings.
- Dashboard pulls live leave balance & history from the GET webhook (8b).
- Apply Leave triggers the human-in-the-loop approval chain (8a).
- HR Assistant is the guarded AI chatbot — input & output guardrails (8c).



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Lunch Break

1 hour

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07

TOPIC 7

Mini Capstone Project

Design · Build · Present

Everything from Activities 1–8, your problem

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TOPIC 7 · MINI CAPSTONE

Your Build — What It Must Include

In small groups, design and build an end-to-end automation for a problem you choose.

1 A trigger

Form, webhook, Telegram or schedule — who or what starts your process?

2 An AI agent

With at least one tool or a RAG source — the judgement in the middle.

3 External data or storage

An API you call, or storage you write — Sheets, a vector DB, a Data Table.

4 A control

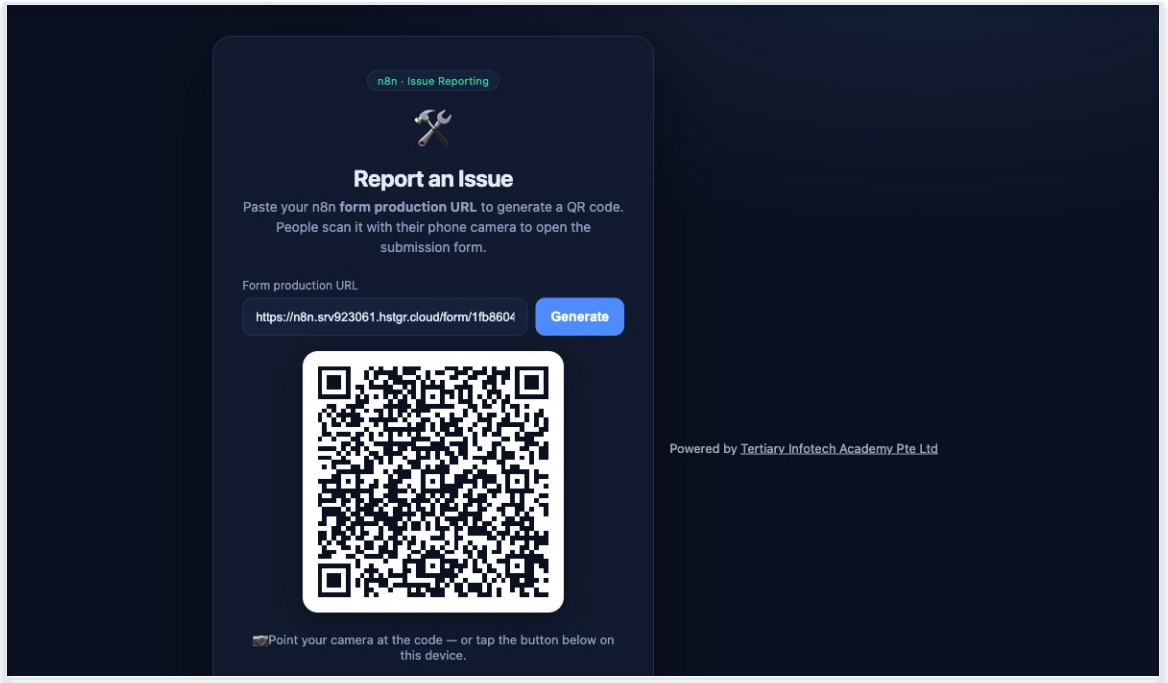
A guardrail or a human-in-the-loop step — decide where the human belongs and defend it.

Deliverables: a working workflow, a short demo, and a 3–5 minute presentation.

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Capstone Example — Issue Reporting

- Paste your n8n Form production URL to generate a QR code.
- People scan the QR with a phone to open the submission form.
- Submissions (with photos) flow into Postgres and a public gallery.
- A complete worked capstone you can adapt for your own project.



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WRAP-UP

Presentation & Assessment

- Present the problem, your design, and what you'd improve.
- Written Assessment (SAQ) — 1 hour · Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials.
- Remember to take the Assessment digital attendance (TRAQOM).

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FINAL ASSESSMENT

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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Thank You!

Keep your local n8n running and keep building your own agents.

Download all flows: github.com/tertiarycourses/TGS-2023035977-Agentic-AI-Automation-with-n8n

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